

ALKALOIDS

Objectives

On completion of this chapter, the student will be able to:

- Describe alkaloids
- Discuss general extraction methods of alkaloids
- Classify alkaloids based on different criteria
- Discuss the major classes of alkaloids including their sources, biological (both medicinal and toxicological) activities.

Contents

- Definition
- Properties
- Distribution and occurrence
- Chemical tests
- Extraction procedures
- Classification
 - Source, chemistry and pharmacology of each class of alkaloid will be discussed

1. Definition

- A precise definition of the term ‘alkaloid’ (alkali-like) is somewhat difficult because there is no clear-cut boundary between alkaloids and naturally occurring complex amines.
- Prior to 19 century, all compounds purified from plants – such as tartaric acid, oxalic acid, and tannins – exhibited acidic properties.
- Potash extracted from burnt wood was later found to contain basic compounds of pharmacological interest.
- *Meissner* suggested in 1819 that these compounds be referred to as alkaloids, meaning a plant-derived substance that displays alkaline (basic) properties.

- Still the definition is not inclusive.
 - E. Winterstein and G. Tiet in 1910 included a five part definition stating
(1) with nitrogen atoms bound in heterocyclic fashion,
(2) with greater or lesser degree of basic character,
(3) marked physiological effects,
(4) complicated molecular structure,
(5) which are found in plants, and, with a few exceptions, are characteristic for particular plant families, genera, or species”

The majority of alkaloids fit this five part definition; however, a number of exceptions exist.

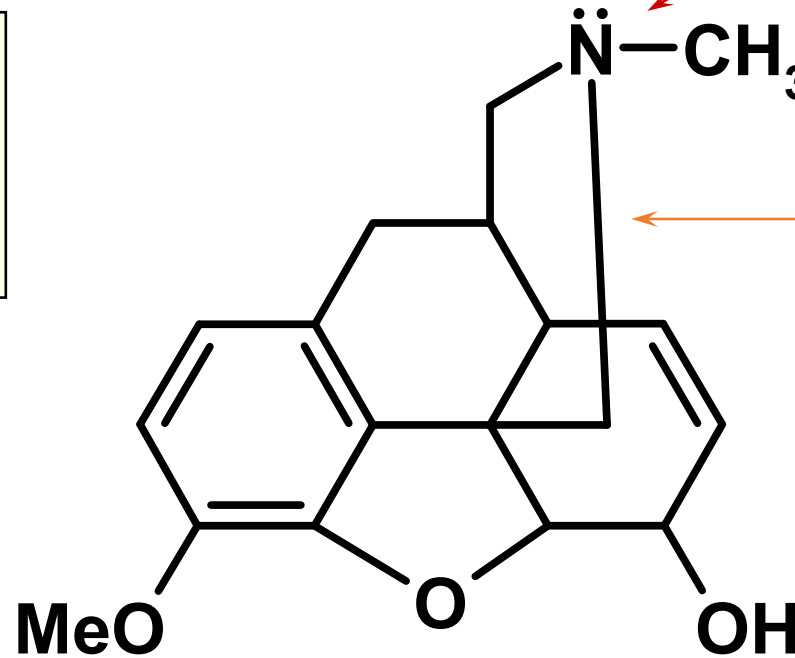
codeine- A TYPICAL ALKALOID

✓ 5-characteristics

basic due to the
unshared pair

contains nitrogen

Plant source.
Most alkaloids
are found in
plants.



heterocyclic ring

Morphin--was the
first alkaloid
discovered
(1804, Serturmer).

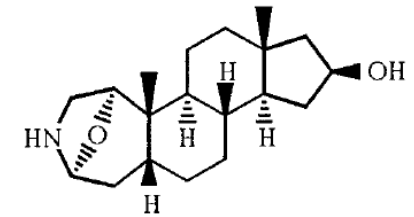
Found only in the Opium Poppy - *Papaver somniferum*

..... not ubiquitous.

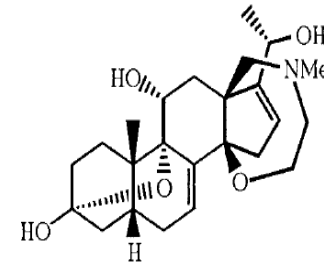
Deviation from Definition

Origin

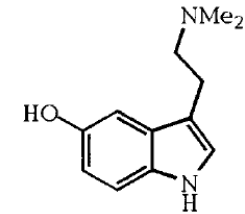
- Samandarine isolated from the skin glands of the European fire salamander exhibit the usual properties of an alkaloid substance, but do not fit the definition of a “true alkaloid” owing to their animal origin.
- Other alkaloids from animal origin: batrachotoxinin A, bufotenine, (–)-castoramine, muscopyridine



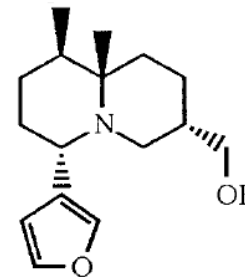
Samandarine



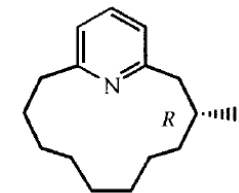
Batrachotoxinin A



Bufotenine



Castoramine

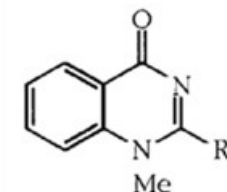


Muscopyridine

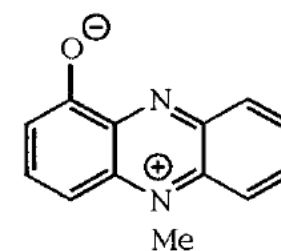
- Alkaloids have also been identified from arthropod, bacterial, and fungal origins.

For example,

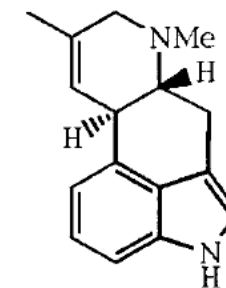
- the quinazole alkaloids, glomerine and homoglomerine discharged from the dorsal glands of the European millipede (*Glomeris marginata*),
- the deep-blue colored alkaloid pyocyanine, isolated from the bacterium *Pseudomonas aeruginosa*, and
- agroclavine, produced by the fungi *Claviceps purpurea* and *Aspergillus fumigatus*.



R = Me; Glomerine
R = Et; Homoglomerine



Pyocyanine



Agroclavine

Nitrogen and basicity

- Other examples of alkaloids exist which also do not adhere to the criteria stipulated in the five part definition of an alkaloid.

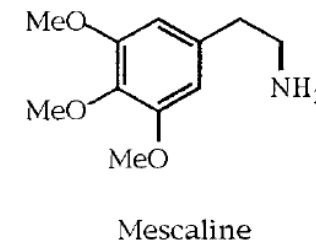
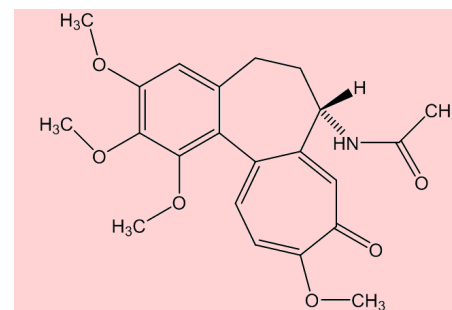
For example,

- the alkaloids colchicines (autumn crocus, *Colchicum autumnale* L.) and mescaline (*Lophophora williamsii*) do not possess nitrogen as part of a heterocyclic system.
- Also **colchicine** is essentially neutral (not basic) and, therefore, does not conform to the original definition of an alkaloid.

The **more available the electrons** the more readily **they can be donated to form a new bond to the proton** and the **stronger base** (Basicity increases with increasing negative charge on nitrogen)

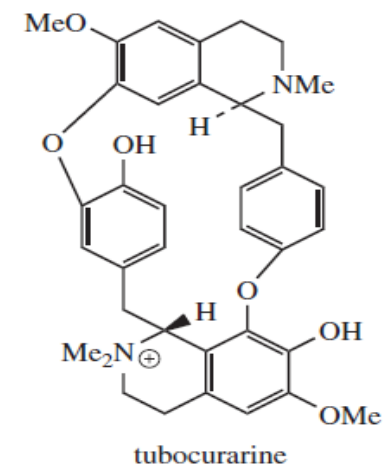
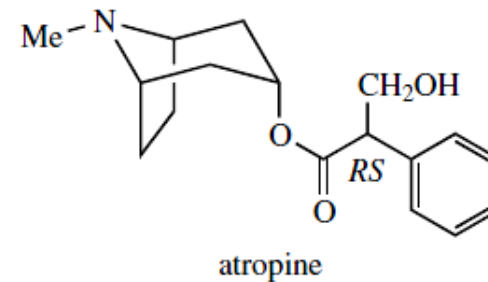
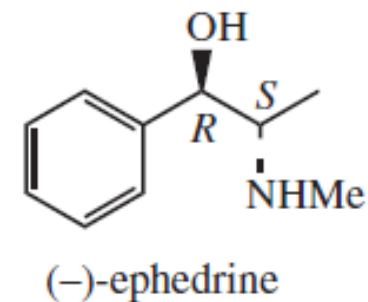
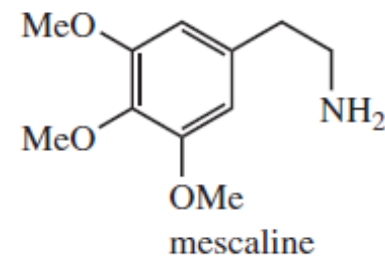
Resonance stabilization decreases basicity

Electron withdrawing groups decrease amine basicity



- Pelletier (1983), however, provides a reasonable summary of an alkaloid's properties as being an
 - “alicyclic compound containing nitrogen in a negative oxidation state which is of limited distribution among living organisms”.
- This definition evidently encompasses all compounds which have hitherto been regarded as alkaloids, but excludes such nitrogenous compounds as simple amines, amino acids, peptides, proteins, nucleic acids, nucleotides, porphyrins, vitamins and nitroso compounds.
- Thus, “Alkaloids are nitrogen-containing organic substances of natural origin with a greater or lesser degree of basic character.”

- Alkaloids usually contain one nitrogen atom, although some like ergotamine may contain up to five.
- The nitrogen may exist:
 - as a primary amine (RNH_2 ; e.g. mescaline),
 - as a secondary amine (R_2NH ; e.g. ephedrine),
 - as tertiary amine (R_3N ; e.g. atropine) or
 - quaternary ($\text{R}_4\text{N}^+\text{X}^-$; e.g. one of the N atoms of tubocurarine).



2. Nomenclature

- Trivial names should end by "**ine**"
- The names of alkaloids are obtained by various ways:
 - ❑ From the generic name of the plant yielding them (atropine from *Atropa belladonna*, hydrastine from *Hydrastis canadensis*)
 - ❑ From the specific name of the plant yielding them (cocaine from *Erythroxylum coca*, belladonnine from *A. belladonna*)
 - ❑ From the common name of the drug yielding them (ergotamine from Ergot)
 - ❑ From the physiologic activity (emetine (from Greek emetikos, "an emetic"))
 - ❑ Occasionally from the discover (pelletierine after Pelletier)
 - ❑ From the physical property (hygrine- hygro = moist)

Sometimes, a suffix (such as “-idine,” “-anine,” “-aline,” “-inine,”; for example quinine, quinidine, hydroquinine) or prefix (such as “iso-”, “pseudo-”, “neo-”, “epi-” and “nor-”) is added to the name of a principal alkaloid to designate another alkaloid from the same source.

Prefixes:

- "Nor-" designates N-demethylation or N-demethoxylation
e.g. norpseudoephedrine and nornicotine
- "Apo-" designates dehydration
e.g. apomorphine
- "Iso-, pseudo-, neo-, and epi-" indicate types of isomers

Suffixes:

- "-dine" designates isomerism as in case of cinchona alkaloids
 - quinidine and cinchonidine are the optical isomers of quinine and cinchonine respectively
- "-ine" indicates, in case of ergot alkaloids, a lower pharmacological activity e.g. ergotaminine is less potent than ergotamine

3. Distribution and occurrence

- May be of plant, animal, insect or microbial origin
- Rare in lower plants
- Dicots are more rich in alkaloids than Monocots
- Families rich in Alkaloids:
 - Apocynaceae, Rubiaceae, Solanaceae and Papaveracea.
- Families free from Alkaloids: Rosaceae, Labiatae

Location in the plant

Alkaloids may occur in every part of a plant, but usually one or more organs have higher content than others.

- All Parts e.g. Datura
- Barks e.g. Cinchona
- Seeds e.g. Nux vomica
- Roots e.g. Aconite
- Fruits e.g. Black pepper
- Leaves e.g. Tobacco
- Latex e.g. Opium

Function of alkaloids in plants

- They may act as protective against insects and herbivores due to their bitterness and toxicity.
- They are, in certain cases, the final products of detoxification in metabolic reactions, therefore considered as waste products of metabolism.
- They may provide nitrogen to the plant organs in case of nitrogen deficiency (source of nitrogen).
- They, sometimes, act as growth regulators in certain metabolic systems.
- They may be utilized as a source of energy in case of deficiency in carbon dioxide assimilation, especially those alkaloids containing a sugar moiety.

Some alkaloids are extremely poisonous e.g.:

- Ergot alkaloids caused epidemic poisoning in the Middle Ages in Europe as a result of feeding on rye bread contaminated with the fungus.
- The extracts of plants containing such alkaloids have long been used as arrow poisons in hunting and warfare e.g. curare extract that contains tubocurarine alkaloid.
- Certain were employed in this respect, as a draught for execution e.g. Socrate's execution, in ancient Greece, with hemlock which contains coniine
- At the time of the Roman Empire, Belladonna (the source of atropine) has been mixed with food with the purpose of murdering.
- Cleopatra, the queen of Egypt used Egyptian henbane (*Hyoscyamus muticus*) that contains hyoscyamine, for suicidal purpose

4. Forms of Alkaloids

- Free bases
- Usually in salts form
 - with organic acids e.g. Oxalic, acetic acids
 - with inorganic acids e.g. HCl, H₂SO₄.
 - with special acids:
e.g. Meconic acid in Opium
Quinic acid in Cinchona
- Glycosidal form with sugars resulting the so called gluco-alkaloids e.g. Solanine in Solanum

5. Physical properties

I- Condition

- Most alkaloids are crystalline solids
- Few alkaloids are amorphous solids e.g. emetine.
- Some are liquids that are either:
 - Volatile e.g. nicotine and coniine, or
 - Non-volatile e.g. pilocarpine and hyoscine

II- Color

- The majority of alkaloids are colorless but some are colored
e.g. Colchicine and berberine are yellow
Canadine is orange
The salts of sanguinarine are copper-red

III- Solubility

- Both alkaloidal bases and their salts are soluble in alcohol
- Generally, the bases are soluble in organic solvents and insoluble in water

Exceptions:

- Bases soluble in water: caffeine, ephedrine, codeine, colchicine, pilocarpine and quaternary ammonium bases
- Bases insoluble or sparingly soluble in certain organic solvents: morphine in ether, theobromine and theophylline in benzene
- Salts are usually soluble in water and insoluble or sparingly soluble in organic solvents

Exceptions:

- Salts insoluble in water: quinine monosulphate
- Salts soluble in organic solvents: lobeline and apotropine hydrochlorides are soluble in chloroform

IV- Optical activity

- Optically active isomers may show different physiological activities
- Usually the l(-) isomer is active than the d(+) isomer
- L-ephedrine is 3.5 times more active than D-ephedrine (relative pressor activities)
- L-ergotamine is 3-4 times more active than D-ergotamine

Exceptions

- D- Tubocurarine is more active than the corresponding L- form
- Quinine (L-form) is antimalarial and its D- isomer quinidine is antiarrhythmic
- The racemic (optically inactive) *dl*-atropine is physiologically active

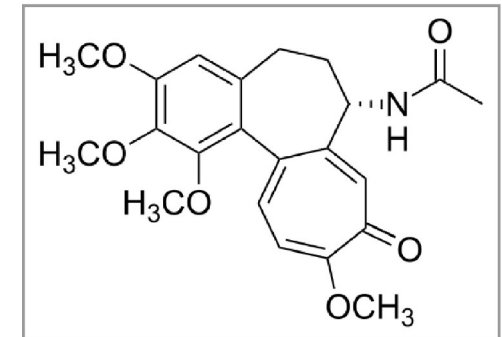
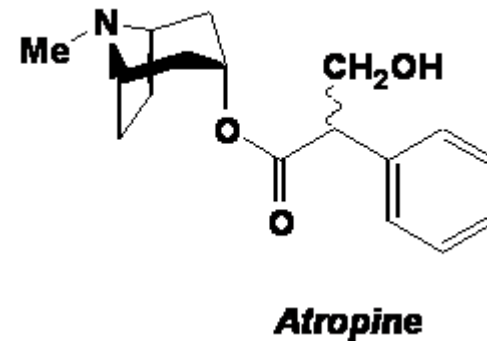
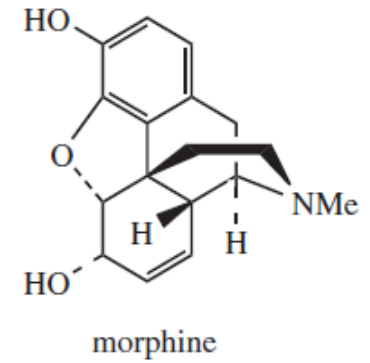
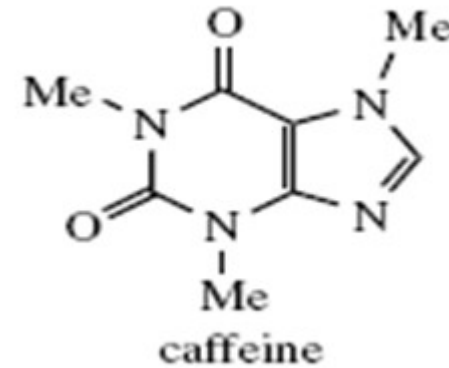
6. Chemical properties

Basicity

- Depends entirely on the availability of the lone pair of electrons on the nitrogen atom
 - Electron withdrawing groups (C=O,) in close proximity to the nitrogen atom decrease the basicity,
 - Electron donating groups (alkyl) enhance it

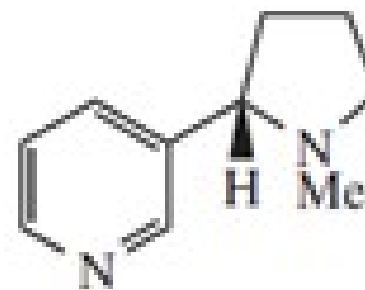
According to basicity Alkaloids are classified into:

- Weak bases e.g. Caffeine
 - Strong bases e.g. Atropine
 - Amphoteric
Phenolic Alkaloids e.g. Morphine
- *Alkaloids with Carboxylic groups
e.g. Narceine
- Neutral alkaloids e.g. Colchicine

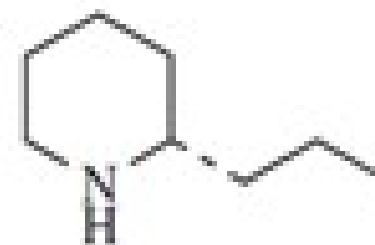


Presence of Oxygen

- Most alkaloids contain Oxygen and are solid in nature e.g. Atropine.
- Some alkaloids are free from Oxygen and are mostly liquids e.g. Nicotine, Coniine



nicotine



(+)-coniine

Factors that affect alkaloidal stability

▪ Heat

Alkaloids are decomposed by heat, except caffeine (sublimable).

▪ Acids

Dehydration:

Some alkaloids lose water molecule to form the anhydro-or apo alkaloids

e.g. Morphine to Apomorphine,
Atropine to Apotatropine.

Demethylation:

N -or O-demethylation of certain alkaloids can take place upon heating with mineral acids

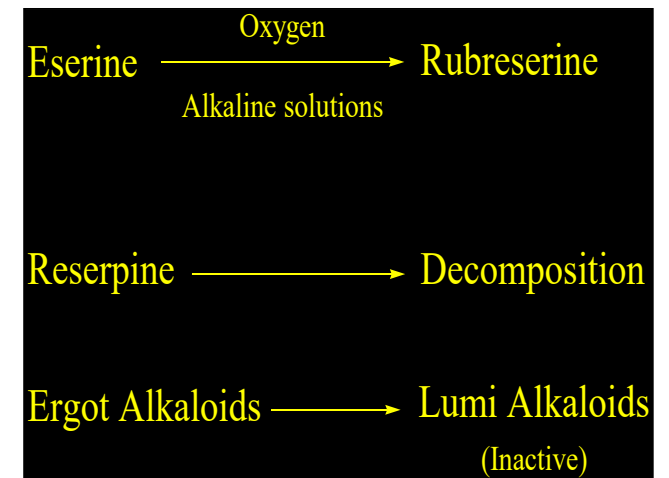
e.g.: Quinine, Narcotine, Codeine, and Papaverine.

▪ Alkali

- **Weak alkalis** e.g. NH_3
 - liberate most alkaloids from their salts
 - can form salts with alkaloids containing a carboxylic group e.g. narceine
- **Strong alkalis**
 - such as NaOH and KOH form salts with phenolic alkaloids
 - cause hydrolysis of ester alkaloids and amide alkaloids : atropine, cocaine and physostigmine

▪ Light and Oxygen

- Some alkaloids are unstable when exposed to light and Oxygen



7. Detection

Tests for detection and identification

Name of reagent	Composition	Remarks
Alkaloidal precipitants:		Color of precipitate:
1.Mayer's	Potassium-mercuric iodide	Creamy white (positive with most alkaloids, except caffeine and dilute ephedrine).
2. Wagner's	Iodine in potassium iodide	Reddish brown
3. Hager's	Saturated solution of picric acid	Yellow
4.Dragendorff's	Potassium bismuth iodide	Orange-reddish brown
5. Marmé's	Potassium cadmium iodide	Yellow precipitate

A negative response,

- No precipitate or turbidity can be taken to mean absence of alkaloids, but
- A positive response

may or may not be due to the presence of alkaloids, as proteins, purines, ammonium salts, lignans, some coumarins

- Therefore, it is always desired to rule out the possibility of a false-test by alkalifying the acidic solution with dilute ammonium hydroxide and subsequently extracting the liberated alkaloid with chloroform.
- The residue thus obtained, after the removal of the solvent (chloroform), is tested with the alkaloid-precipitating reagents.
- Now, if the test is positive, the presence of an alkaloid is almost confirmed

8. Extraction and isolation

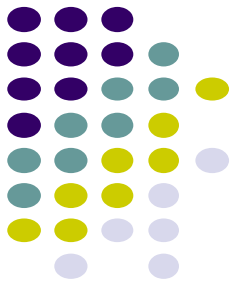
- The general procedures for the isolation of alkaloids are largely based on
 - the alkaline nature of most alkaloids,
 - the subsequent ability to form salts with acids and on the relative solubilities of the alkaloid bases and their salts in water and in various organic solvents
- Sometimes defatting the plant material with petroleum ether or hexane may be necessary to remove lipophilic materials (fats, waxes, terpenes, pigments) which may interfere with the extraction procedure, for example, by causing the formation of emulsion.

Method 1

- The powdered material that contains alkaloidal salts is moistened with alkaline substances like sodium bicarbonate, ammonia, calcium hydroxide, etc., which combines with acids, tannins and other phenolic substances and sets free the alkaloids bases.
- Extraction is then carried out with organic solvents such as ether or petroleum spirit.
- The concentrated organic liquid is then shaken with aqueous acid and allowed to separate.
- Alkaloid salts will be present in aqueous liquid, while many impurities remain behind in the organic liquid.

Method 2

- The collected powdered material is extracted with water or aqueous alcohol containing dilute acid.
- Chloroform or other organic solvents are added and shaken to remove the pigments and other unwanted materials.
- The free alkaloids are then precipitated by the addition of excess alkalis like, sodium bicarbonate or ammonia and separated by filtration or by extraction with organic solvents.

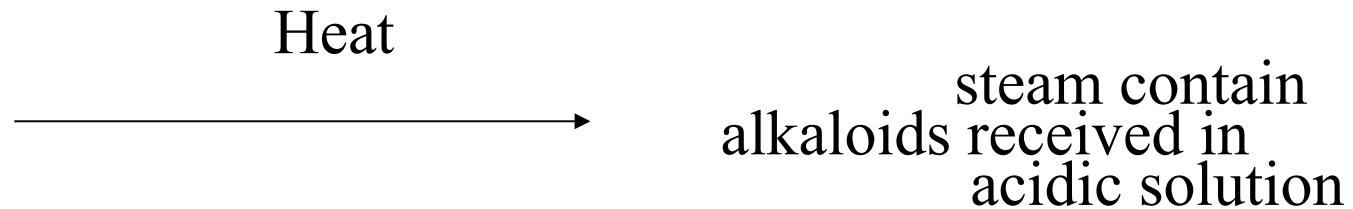


Volatile Alkaloids

- The best way for their extraction is steam distillation.

e.g. Nicotine

- Plant material + water + Fixed alkali

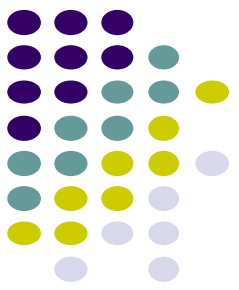


Choice of alkalis

- Alkalis are used to liberate free bases. Alkalis must be strong enough to liberate free bases.
- It has been observed that the salt of a *strongly basic alkaloid* with a mineral acid usually tends to undergo cleavage under the influences of a stronger base. Likewise, the corresponding salt of a *weakly basic alkaloid* and a relatively weak organic acid shall require a rather weaker base for its cleavage.

choice of strong alkalis must be avoided in some cases:

- 1- Ester Alkaloids e.g. Solanaceous Alkaloids
- 2- Amide Alkaloids e.g. Colchicine
- 3- Phenolic Alkaloids e.g. Morphine
- 4- Lactone Alkaloids e.g. Pilocarpine
- 5- Fatty Drugs due to saponification and emulsion formation.



NH_4OH :

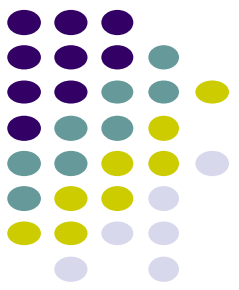


Most widely used due to many advantages:

- Strong enough to liberate most of alkaloids from their salts
- Safe
- Volatile so easy to get rid of it

Other Alkalis:





9. Classification of Alkaloids

a) **Taxonomic classification**

- ✓ based on their respective distribution in a variety of taxa.

b) **Pharmacological classification**

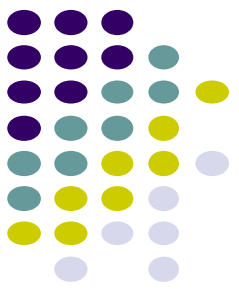
- ✓ specific pharmacological characteristics, not quite common

c) **Chemical classification**

- ✓ presence of the basic heterocyclic nucleus, type of nitrogen and type of ring structure
- ✓ most widely accepted and common mode of classification of alkaloids

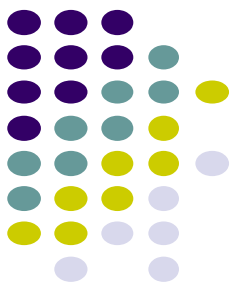
d) **Biosynthetic classification**

- Based on the biogenetic pathway and precursor from which the alkaloids in question are produced



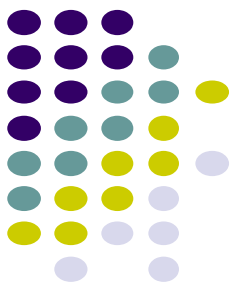
According to Hegnauer's classification which is based on both the type of nitrogen and the biochemical origin: 3 types

- **True alkaloids:** are derived from amino acids and they share a heterocyclic ring with nitrogen.
 - These alkaloids are highly reactive substances with biological activity even in low doses. e.g. Atropine
- **Protoalkaloids:** are derived from amino acids and do not have nitrogen in a heterocyclic ring. e.g. Ephedrine
- **Pseudo alkaloids:** are not derived from amino acids but have nitrogen in a heterocyclic ring e.g. Caffeine
 - most often have all of the characteristics of the true alkaloids, but they are not derived from amino acids. Their nitrogen derives from the formal incorporation of ammonia (i.e. via amination and transamination) into a carbon skeleton



Based on pharmacological action

Analgesics	e.g. Morphine and Codeine
CNS stimulants	e.g. Caffeine and Strychnine
Anti-cancers	e.g. Vincristine, Vinblastine and Taxol
Mydriatics	e.g. Atropine
Myotics	e.g. Pilocarpine
Anti-asthmatics	e.g. Ephedrine
Anti-tussives	e.g. Codeine
Expectorants	e.g. Lobelline
Anti-hypertensives	e.g. Reserpine
Smooth muscle relaxants	e.g. Atropine and Papaverine
Skeletal muscle relaxants	e.g. Tubocurarine
Anthelmintics	e.g. Pelletierine
Antiparasitics	e.g. Quinine and Emetine



Based on chemical structure

A-types of nitrogen, Heterocyclic or non- heterocyclic

B- according to type of ring structure.

a) Non-Heterocyclic or atypical alkaloids

*** Sometimes called Protoalkaloids or Biological amines e.g. Ephedrine, Colchicine, and Taxol.**

*** All have exocyclic N-atoms.**

b) Heterocyclic or typical alkaloids which subdivided into several groups according to their ring structure.



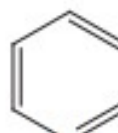
Basic alkaloid skeletal structures in natural products



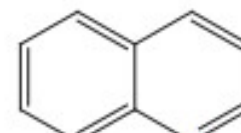
Pyrrole



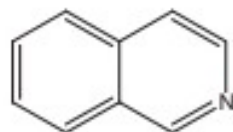
Pyrrolidine



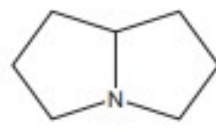
Pyridine



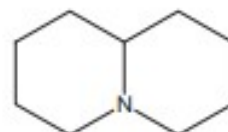
Quinolone



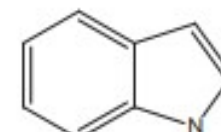
Isoquinoline



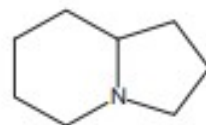
Pyrrolizidine



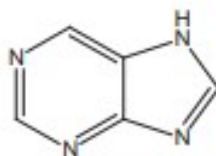
Quinolizidine



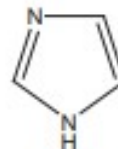
Indole



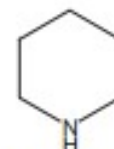
Indolizidine



Purine



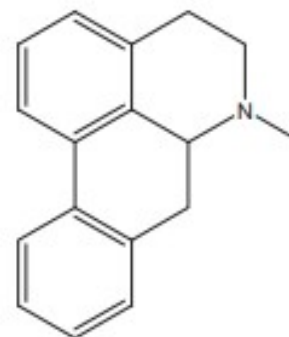
Imidazole



Piperidine

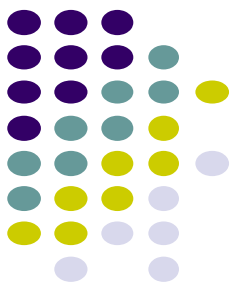


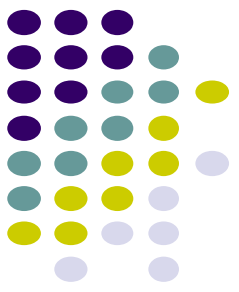
Tropane



Aporphine

- i. Pyrrolidine alkaloids e.g., Hygrine;
- ii. Piperidine alkaloids e.g., Lobeline;
- iii. Pyrrolizidine alkaloids e.g., Senecionine;
- iv. Tropane alkaloids e.g., Atropine;
- v. Quinoline alkaloids e.g., Quinine;
- vi. Isoquinoline alkaloids e.g., Morphine;
- vii. Aporphine alkaloids e.g., Boldine;
- viii. Indole alkaloids e.g., Ergometrine;
- ix. Imidazole alkaloids e.g., Pilocarpine;
- x. Diazocin alkaloids e.g., Lupanine;
- xi. Purine alkaloids e.g., Caffeine;
- xii. Steroidal alkaloids e.g., Solanidine;
- xiii. Amino alkaloids e.g., Ephedrine;
- xiv. Diterpene alkaloids e.g., Aconitine

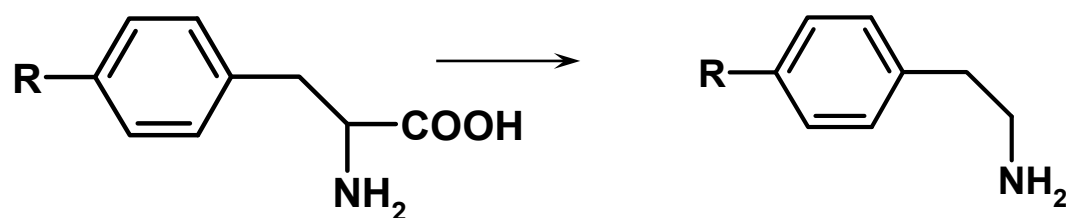
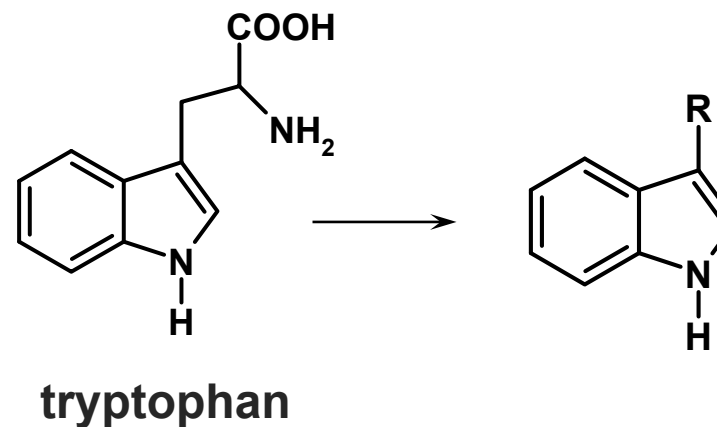
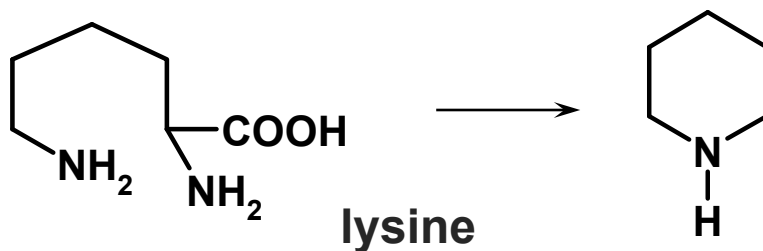
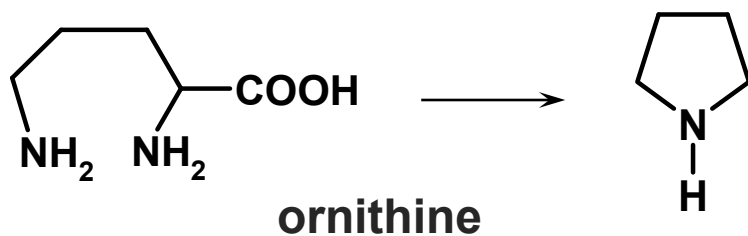




Biosynthesis of Alkaloids

- The biosynthesis of alkaloids starts from the primary metabolites
 - The amino acids or their decarboxylated products
 - Principal amino acids are
 - ✓ ornithine, lysine, nicotinic acid, tyrosine, tryptophan, anthranilic acid, and histidine
- e.g., tryptamine (decarboxylated tryptophan) gives rise to the indole class of alkaloids

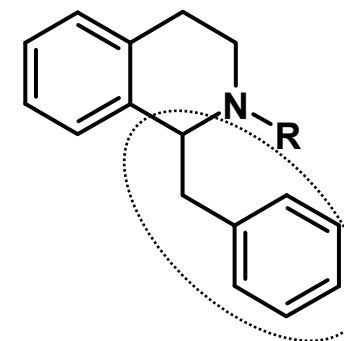
MOST ALKALOIDS ARE DERIVED FROM α -AMINO ACIDS
SOME OF THE MAJOR RELATIONSHIPS ARE SHOWN BELOW



R = H phenylalanine

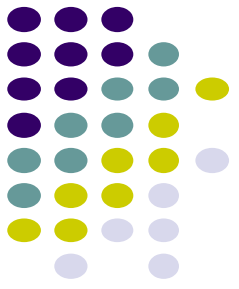
R = OH tyrosine

and

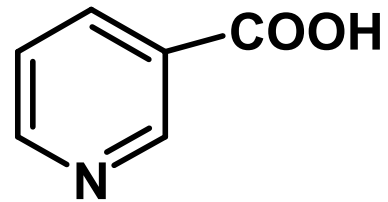


R = H, CH₃
R' = H, alkyl

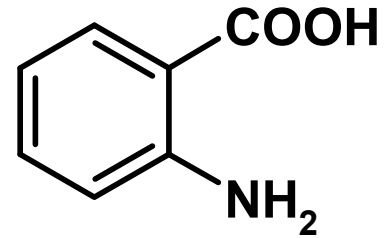
A COUPLE OF THE β -AMINO ACIDS ARE SOMETIMES INVOLVED IN ALKALOID PRODUCTION ALSO



These β -amino acids are :



nicotinic acid



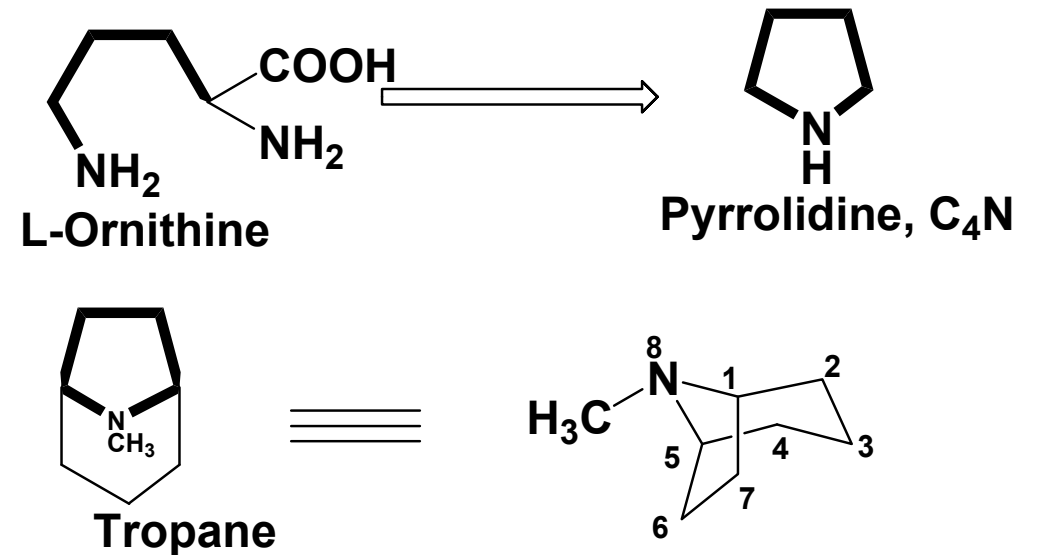
anthranilic acid

1. Ornithine Derived Alkaloids

- ornithine supplies a C₄N building block to the alkaloid, principally as a **pyrrolidine ring** system, but also as part of the tropane alkaloids.

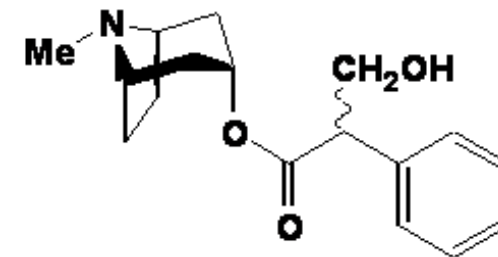
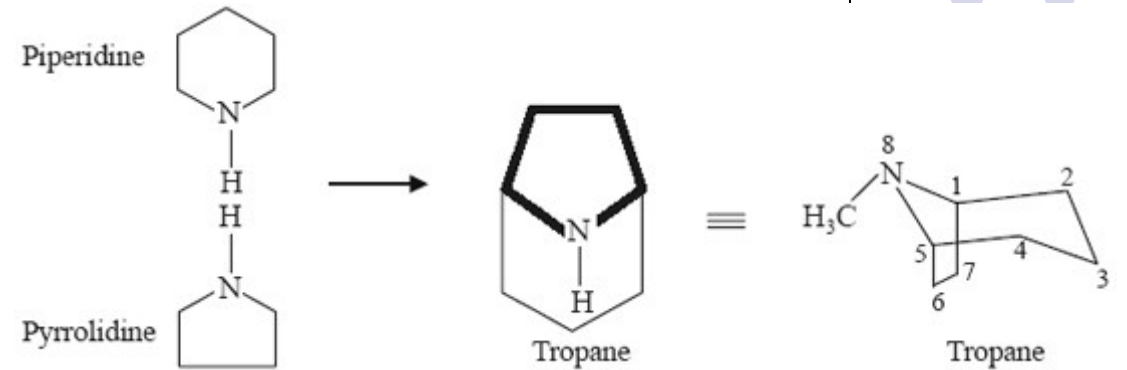
It gives rise/ precursor to:

- Tropane Alkaloids
 - Solanaceous alkaloids
 - Coca alkaloids
- Pyrrolizidine alkaloids

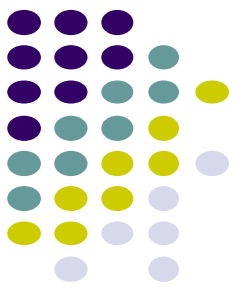


1.1 Tropane Alkaloids

- Tropane is a bicyclic compound formed by condensation of pyrrolidine + piperidine with one nitrogen atom in common
- derived from L-ornithine
- is a seven carbon bicyclic ring, with a nitrogen bridge between carbons 1 and 5.
- It contains two asymmetric carbons at C-1 and C-5 but it is optically inactive due to intermolecular compensation.

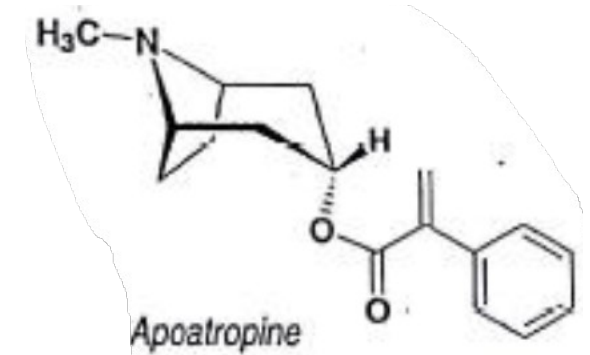
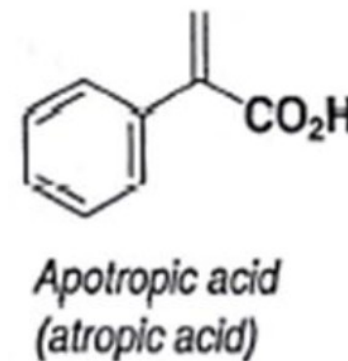
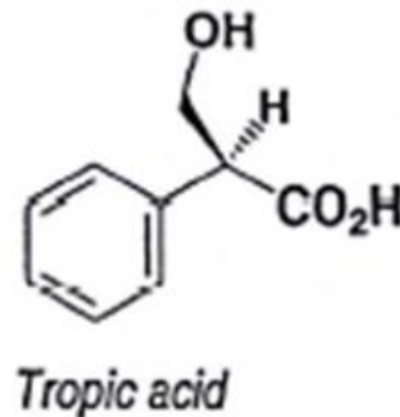
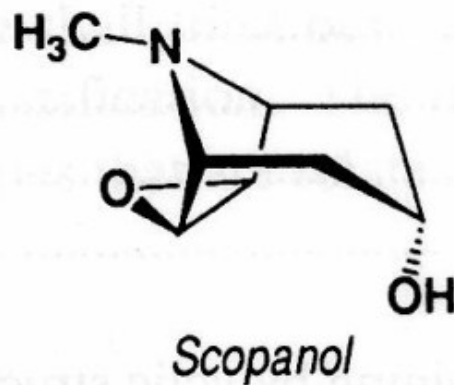
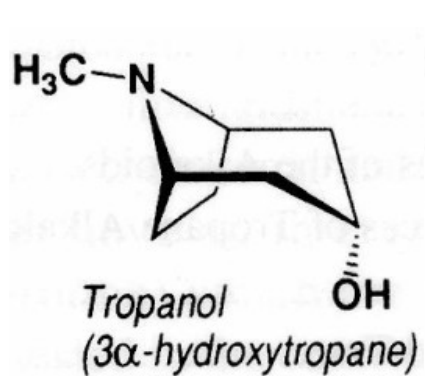


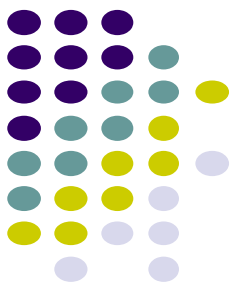
Atropine



- With few exceptions tropane alkaloids are esters of tropane alcohols and of acids of various structures either aliphatic or aromatic.
- they are unstable towards acids and alkalis and are thermolabile
- Tropane is considered as the parent base of several alkaloids obtained from a number of families including:

Solanaceae	e.g. Atropine, Hyoscyamine, Scopolamine (Hyoscine), Meteloidine and Belladonnine.
Erythroxylaceae	e.g. Cocaine, Cinnamyl cocaine and α -and β -Truxillines.
Convolvulaceae	e.g. Convolvine.
Dioscoreaceae	e.g. Dioscorine.





1.1.1. Solanaceous Tropane Alkaloids

- **Occurrence:**

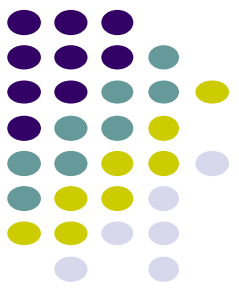
<i>Atropa belladonna</i>	(Belladonna Leaves)
<i>Dature stramonium</i>	(Thornapple)
<i>Hyoscyamus muticus</i>	(Egyptian henbane)
<i>Hyoscyamus niger</i>	(European henbane)

- **Main Alkaloids are:**

A- Atropine.

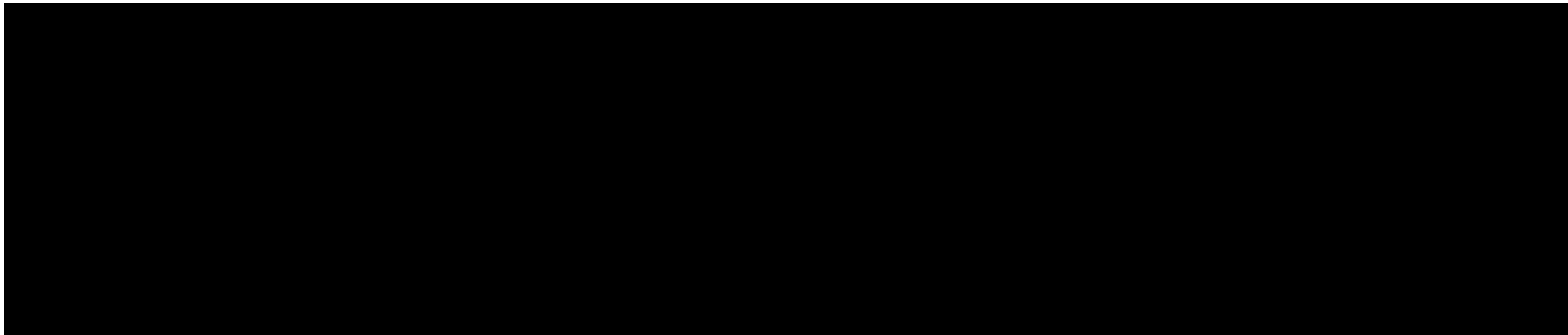
B- Hyoscyamine.

C- Hyoscine (Scopolamine)



A. Atropine & Hyoscyamine

- Atropine is racemate composed of equimolar concentrations of (S)- and (R)-atropine.
- Hyoscyamine is the major natural alkaloid with negative optical rotation (*l*- form) sometimes known as levo-atropine.
- During extraction hyoscyamine racemizes to the optically inactive *dl* Atropine.
- Both alkaloids composed of tropine base and tropic acid.



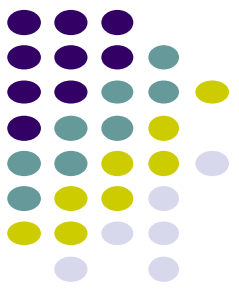


B. Hyoscine (Scopolamine)

- Hyoscine is an ester of *l*-tropic acid with scopolamine base
- Hyoscine is a syrupy liquid

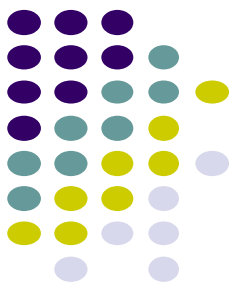


Pharmacological actions and uses



The three Alkaloids are anticholinergic agents:

- Decrease saliva and GIT secretions so used preoperative
- Decrease motility of smooth muscles so used as antispasmodics
- Stimulate respiratory system
- A mydriatic (causes dilatation of the eye pupil).
- An antidote to organophosphorus insecticides, physostigmine or neostigmine (CE inhibitors)
- Hyoscine has more central effect so it is sedative and hypnotic
- Hyoscine is mainly used as antiemetic

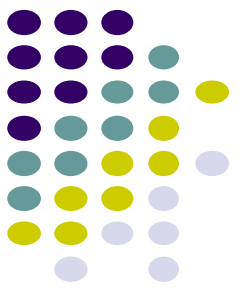


1.1.2. Erythroxyton (Coca) Alkaloids

- Coca leaves (2 %).
- According to their chemical structures,

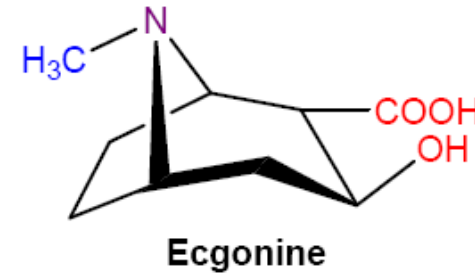
Coca alkaloids are classified into 3 basic types:

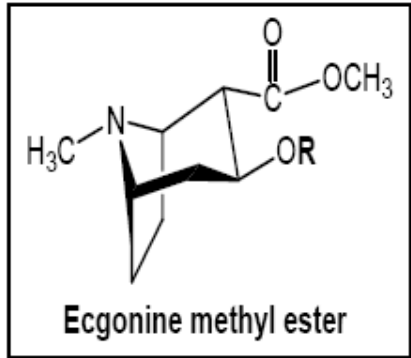
Ecgonine derivatives	e.g. cocaine, cinnamyl cocaine, α - and β - truxillines
Pseudotropine derivatives	e.g. tropacocaine and velerine
Pyrrolidine derivatives	e.g. hygrine and cuscohygrine (the principal non-ester alkaloids of coca leaves).



A. Ecgonine derivatives

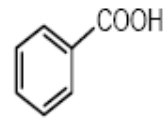
- Ecgonine base contain both acidic and alcoholic groups
- The carboxylic group is esterified, in most cases, with **methyl alcohol**
- The alcoholic group is esterified with **different acids, giving different alkaloids**



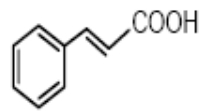


Alkaloid

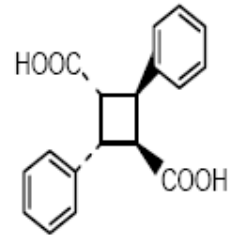
- Cocaine R = Benzoyl
- Cinnamylcocaine R = Cinnamoyl
- Truxilline R = Truxilloyl
- Truxilline R = Truxilloyl



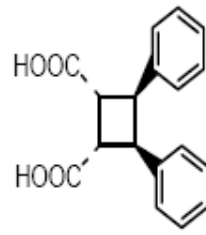
Benzoic acid



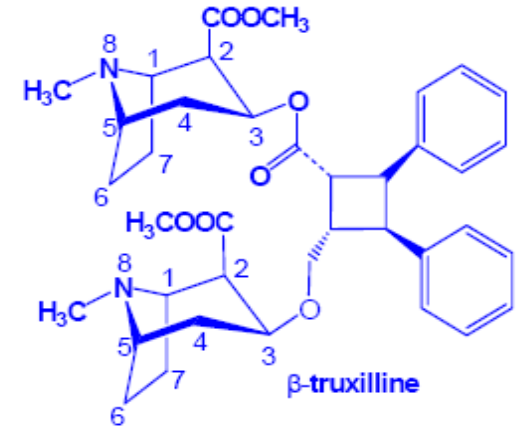
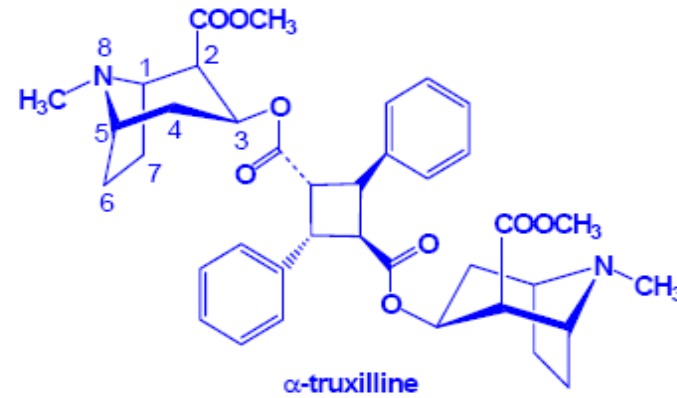
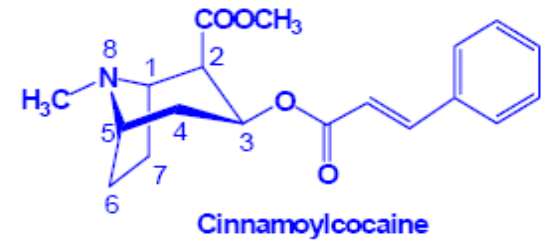
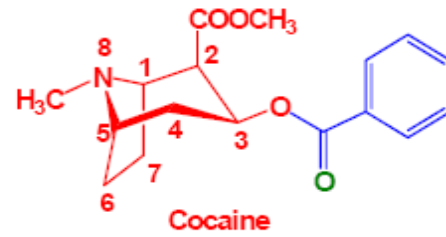
Cinnamic acid



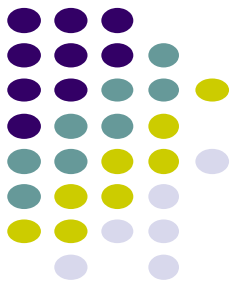
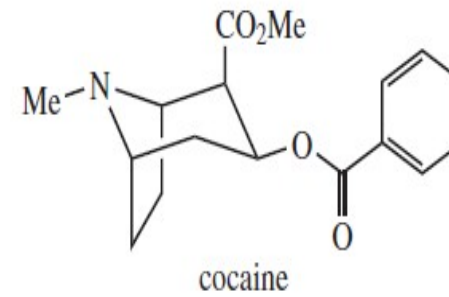
α -Truxillic acid



β -Truxinic acid

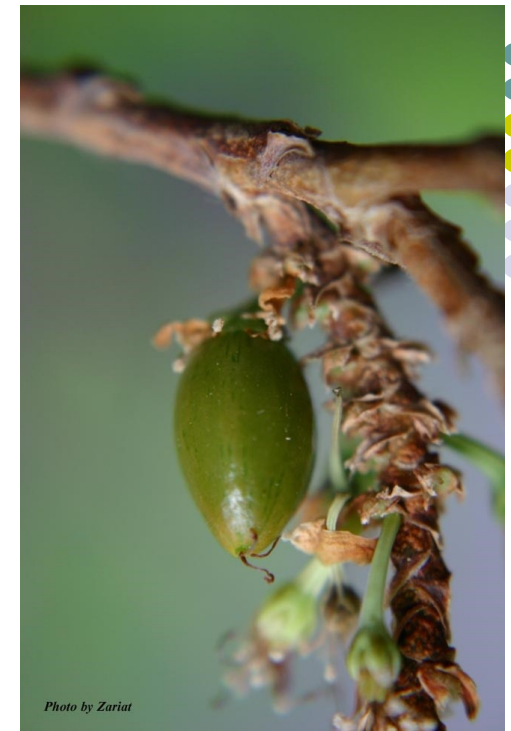


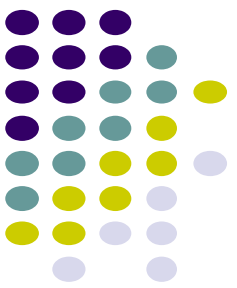
Cocaine



- Cocaine is the benzoic acid ester of the tropane base **methyl ecgonine**.
- Diester Alkaloid and 3° amine (basic) alkaloid
- It is the main component of coca leaves which are obtained from species of *Erythroxylum* (Erythroxylaceae).
- It is the major Alkaloid in Coca leaves
- Leaf is mixed with lime, why??
- Heating at 160 °C in conc. HCl leads to hydrolyses of cocaine to MeOH, Benzoic acid and Ecogonine base
- It is levorotatory, white, crystalline, and slightly volatile
- It causes numbness of the lips and tongue when locally applied

- Cocaine is a highly addictive central stimulant
- Inhibits the re-uptake of the neurotransmitters dopamine and norepinephrine at synapses, and inhibits monoamine oxidase, the enzyme that degrades dopamine, epinephrine, and norepinephrine, so prolonging and augmenting their effects
- This leads to euphoria, hyperactivity, suppression of hunger and fatigue.
- The drug may also be injected intravenously or the vapour inhaled.
- For inhalation, the free base or ‘crack’ is employed to increase volatility.
- The vaporized cocaine is absorbed extremely rapidly and carried to the brain within seconds, speeding up and enhancing the euphoric lift.
- Taken in this form, cocaine has proved highly addictive and dangerous.





Uses

- Used externally as local anesthetic.
- It has a CNS stimulant effect, followed by a hypnotic effect
 - It is **quickly absorbed** through mucous membranes.,
 - 50 mg of Cocaine leads to euphoria and hallucination.
 - Larger doses lead to cerebral cramps, hyperirritabilities and **paralysis**.
- Cocaine is an addictive drug

Toxicity:

- The minimum lethal dose is 1.2 g, but susceptible individuals have died from doses as small as 30 mg when applied to mucous membrane

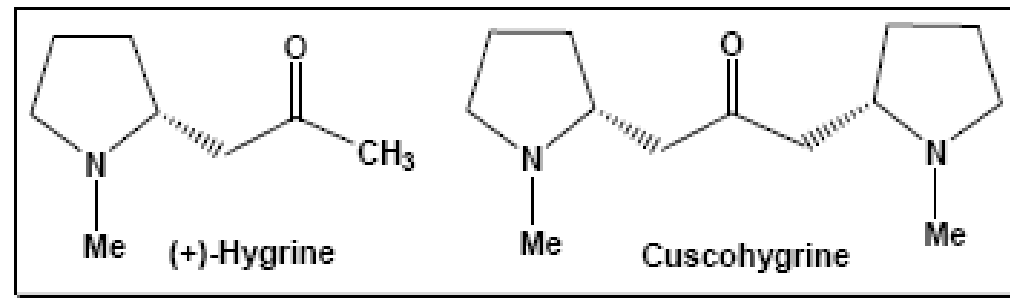
B. Pyrrolidine derivatives

Hygrine

- It is a non-ester strongly basic liquid alkaloid, easily decomposed in air
- Soluble in water and organic solvents
- Hygrine + gold chloride or platinum chloride → Plate of aggregate crystals

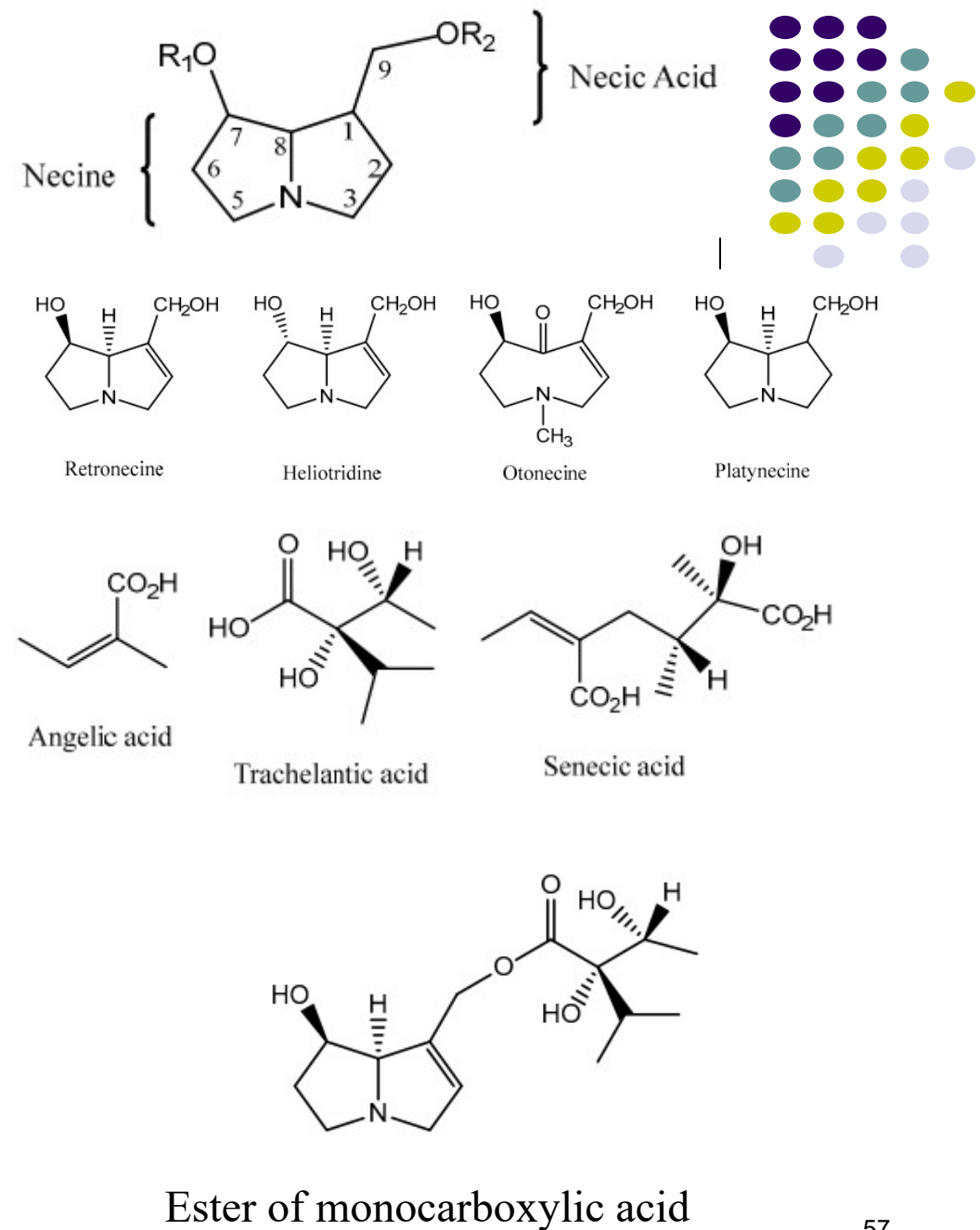
Cuscohygrine

- Is the principal non-ester alkaloid in coca leaves
- It occurs as an oily alkaloid, containing two tertiary nitrogen atoms
- It is miscible with water
- Cuscohygrine + hydrobromide or nitrate salts → Cluster of needle crystals

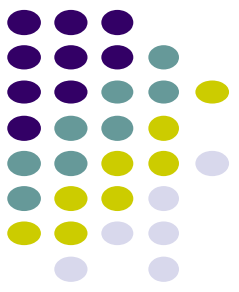


2. Pyrrolizidine Alkaloids

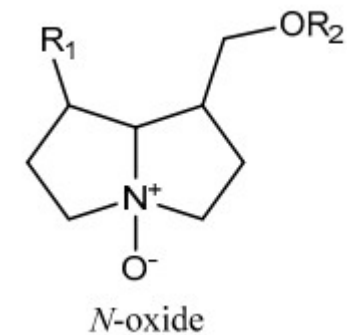
- Two molecules of ornithine are utilized in formation of the bicyclic pyrrolizidine skeleton
- Are the esters of 1-hydroxymethylpyrrolizidines (necines) and necic acids
- They rarely occur in the free form as a pyrrolizidine base, being instead found as esters (mono-, di- or macrocyclic diesters) formed by a necine base (amino alcohols) and one or more necic acids (mono- or dicarboxylic aliphatic acids), which are responsible for their structural diversity .
- Ester linkages are found at positions C9 and C7

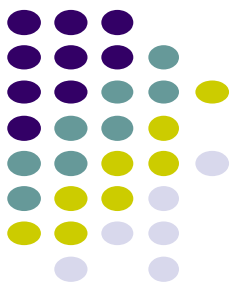


Ester of monocarboxylic acid

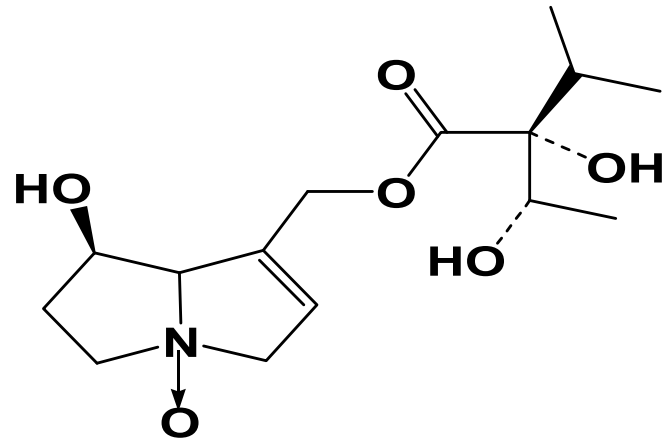


- They are usually found in the form of tertiary bases or pyrrolizidine alkaloids *N*-oxides (PANO)
- The *N*-oxides are easily changed back to the tertiary amines by mild reduction, as will occur in the gut of a herbivore
- Many pyrrolizidine alkaloids are *hepatotoxic*, *mutagenic*, and *carcinogenic*.
- Toxicity is due to formation of reactive pyrrole structures by mammalian liver oxidases
- Pyrrole are potent alkylating agents and react with suitable cell nucleophiles
e.g. nucleic acids and proteins
- Restricted distribution
 - Leguminosae/Fabaceae (e.g. Crotalaria)
 - Boraginaceae (e.g. Heliotropium, Cynoglossum, and Symphytum)
 - Compositae/Asteraceae (e.g. Senecio





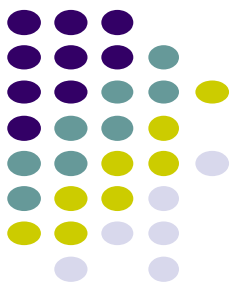
- Some PAs are in lower doses are on clinical trials for the **treatment of cancer**



Indicine-N-oxide

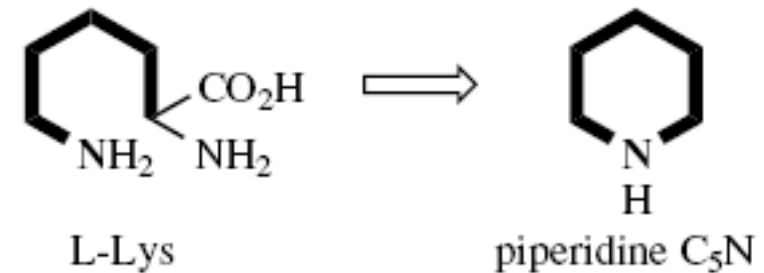
Indicine-*N*-oxide

- from *Heliotropium indicum* (Boraginaceae)
- significant antileukaemic activity in clinical trials
- undesirable hepatotoxicity prevented any further development



2. Lysine derived alkaloids

- Homologue of L-ornithine
- As with ornithine, the carboxyl group is lost, the ϵ -amino nitrogen rather than the α -amino nitrogen is retained, and lysine thus supplies a C₅N building block
- **Lysine gives rise to**
 - Piperidine Alkaloids
 - Quinolizidine Alkaloids
 - Indolizidine alkaloids



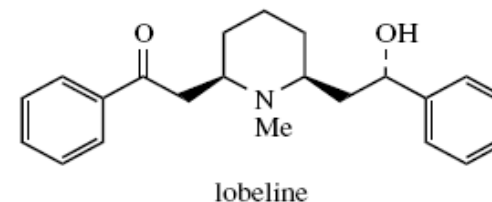
2.1 Piperidine Alkaloids

❑ Lobelia alkaloids

- Lobelia or Indian tobacco consists of the dried leaves and tops of *Lobelia inflata* (Campanulaceae)
- contains 0.2–0.4% of alkaloids (Lobeline is chief)
- The C₆C₂ is derived from Phenylalanine

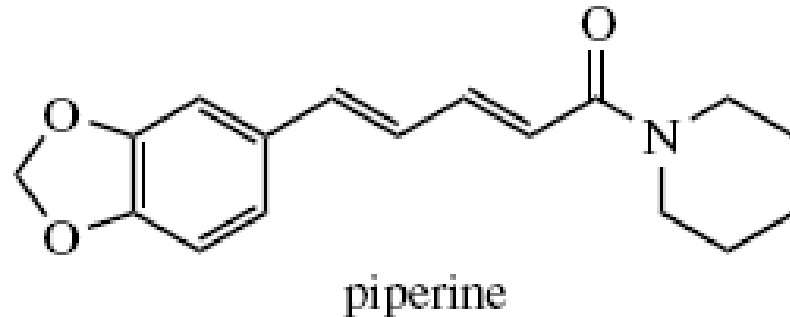
Uses

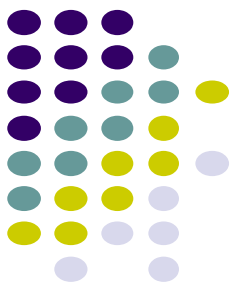
- as smoking deterrent
 - it is found that lobeline stimulates nicotinic receptor sites in a similar way to nicotine, but with a weaker effect.
- used to relieve asthma and bronchitis b/c it stimulates respiration though in large doses it can be quite toxic





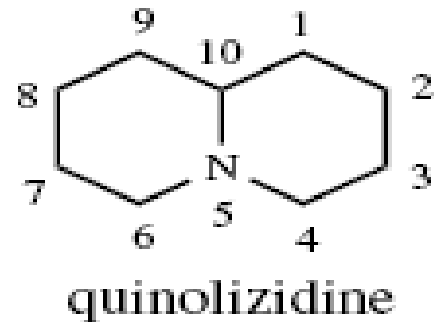
- Alkaloids of black pepper
- The pungency of the fruits of black pepper (*Piper nigrum*; Piperaceae), a widely used condiment, is mainly due to the piperidine alkaloid piperine

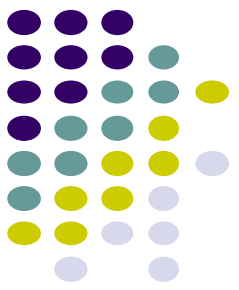




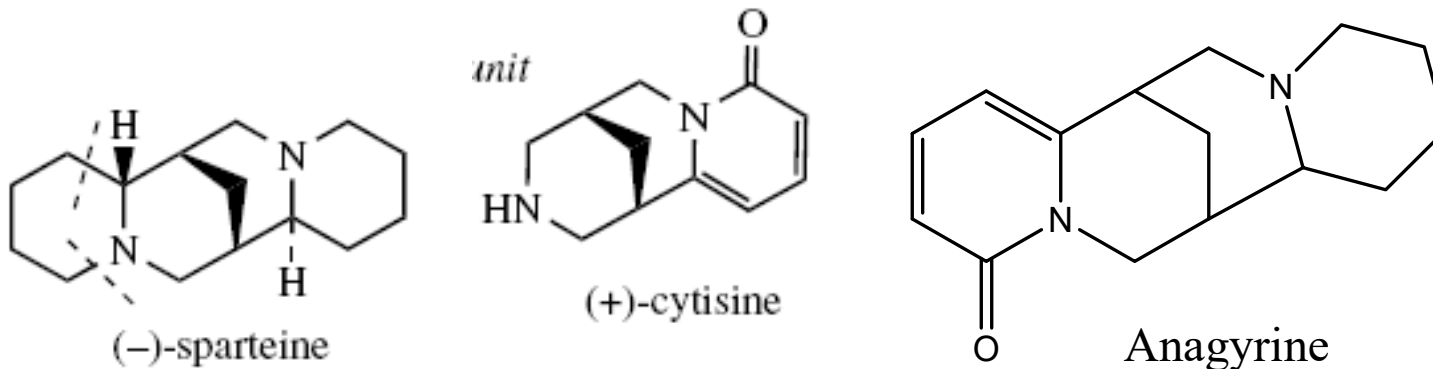
2.2 Quinolizidine Alkaloids

- are mainly found in plants of the Leguminosae/Fabaceae family
- Quinolizidine skeleton is formed from two molecules of lysine
- are also known as **lupine alkaloids**





- are toxic to herbivores by a variety of mechanisms
 - ✓ inhibition of Na^+ and K^+ channels and interaction with nicotinic and muscarinic acetylcholine receptors.
- Anagyrine and cytisine isolated from *Lupinus sericeus* (Fabaceae) are teratogenic in animals

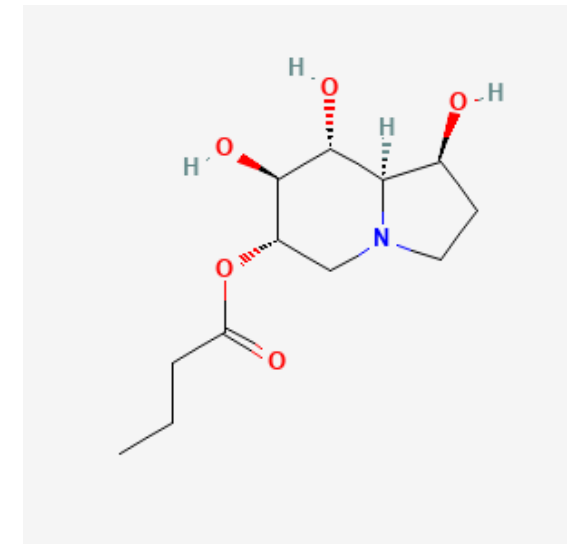
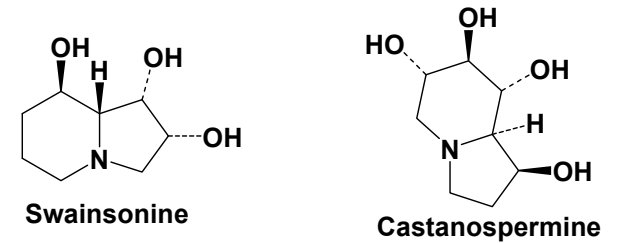
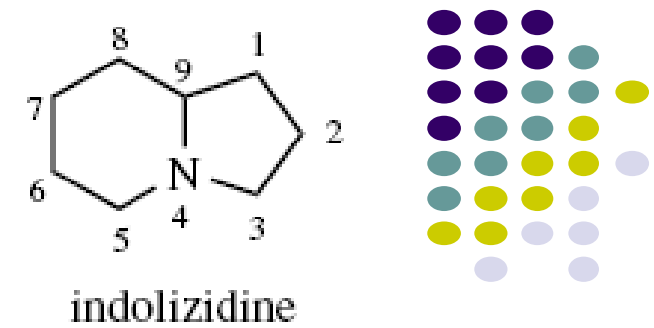


2.3 Indolizidine Alkaloids

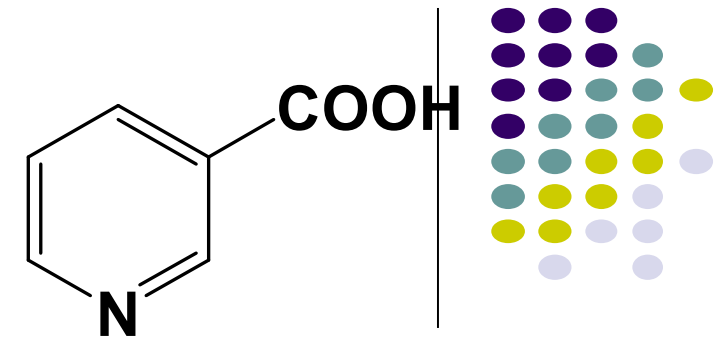
swainsonine - *Swainsona canescens*

castanospermine - *Castanospermum austral*

- have demonstrated activity against the AIDS virus HIV
 - by their ability to inhibit glucosidase enzymes involved in glycoprotein biosynthesis
- The ester 6-O-butanoylcastanospermine (Celgosivir) is currently in clinical trials as an anti-AIDS agent
- are also toxic to animals, causing severe gastro-intestinal upset and malnutrition by severely affecting **intestinal hydrolases**
- found in some fungi (e.g. *Rhizoctonia leguminicola*)



3. Nicotinic Acid Derived Alkaloids



- are commonly known as the ‘**Pyridine Alkaloids**’
- the ‘*pyridine unit*’ has its origins in **vitamin B₃ (nicotinic acid)**
- Nicotinic acid is a non-protein amino acid
- Alkaloids derived from nicotinic acid include
 - Alkaloid of castor beans
 - Alkaloids of areca nuts
 - Alkaloids of tobacco leaves

3.1. Alkaloid of castor beans

- is obtained from *Ricinus communis*
- Ricinine is the sole alkaloid and is not basic
 - mildly toxic
 - Ricinine can be found in all parts of the plant and it is a quite strong insecticide.
- The main toxic principle is a glycoprotein ricin (strychnos and cyanides)
- Ricinine (3-cyano-4-methoxy-N-methyl-2-pyridone) has a CNS stimulant activity, where intoxicated animals by seeds cake showed various symptoms ranging from loss of appetite, diarrhea, weakness, impaired feeling and emotion, and eventually death because of respiratory arrest.



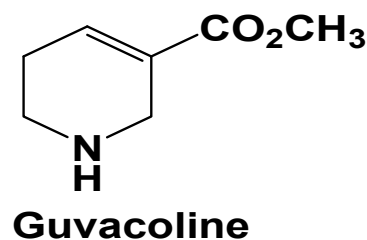
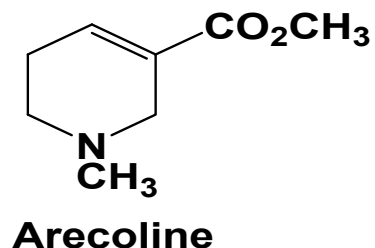
3.2. Areca alkaloids

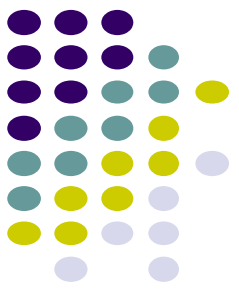
- *Areca catechu* (Palmae/Arecaceae)
- Nuts are mixed with lime, wrapped and then chewed
- The teeth and saliva of chewers *stain bright red*
- Seeds contain 0.2-0.4%- alkaloids & 8-15% tannins
- The major stimulant alkaloid is arecoline occurs together with arecaidine, guvacine and guvacoline



Arecoline is

- an agonist for muscarinic acetylcholine receptors
- employed in veterinary practice as a vermicide to eradicate worms

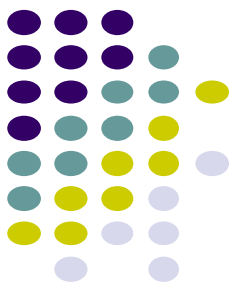




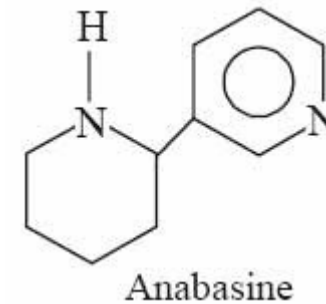
3.3. Alkaloids of tobacco leaves

- Tobacco is the cured and dried leaves of *Nicotiana tabacum* (Solanaceae)
- Tobacco leaves may contain
 - **Volatile alkaloids**
 - 0.6–9% of (–)-nicotine
 - anabasine and nornicotine (small amount)
 - **non-volatile alkaloids**
 - nicotine and nicotyrine
 - **solid alkaloid**
 - nicotelline
- In the leaf, the alkaloids are typically present as salts with malic and citric acids

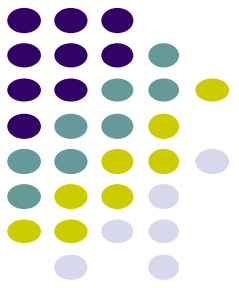




- **Nicotine**
 - a pyridine ring from nicotinic acid
 - a pyrrolidine ring from ornithine
- **Anabasine**
 - a pyridine ring from nicotinic acid
 - a piperidine ring from lysine

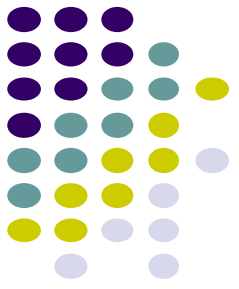


Nicotine

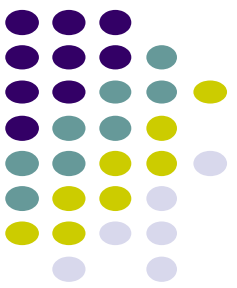


- Is a colorless liquid, which can be prepared from tobacco by steam distillation
- It also occurs in club moses (*Lycopodium* spp) and in higher plants such as *Erythroxylum coca*
- It is optically active and miscible with water, ether and light petroleum
- It possesses two basic nitrogen atoms, the weaker one is that in the pyridine ring, **WHY???**
- Nicotine exerts two effects, a stimulant effect exerted at the locus ceruleus and a reward effect in the limbic system

Uses of nicotine

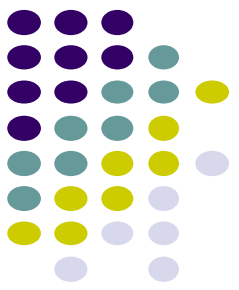


- Nicotine and anabasine are used as an insecticide in the form of sulphate salt
 - ✓ Nicotine is very potent against aphids
- Employed as chewable tablets or lozenges for the treatment of smoking withdrawal syndrome.



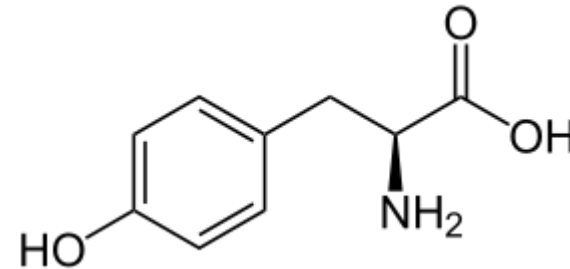
Toxicity

- Tobacco smoke contains a number of highly carcinogenic chemicals formed by incomplete combustion
 - benzpyrene,
 - 2-naphthylamine
 - 4-aminobiphenyl
- Tobacco smoking also contributes to:
 - atherosclerosis
 - chronic bronchitis
 - emphysema
- is regarded as the single most preventable cause of death in modern society

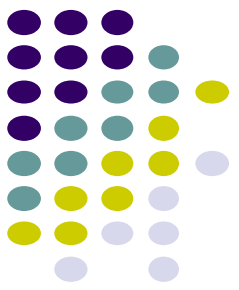


4. Tyrosine Derived Alkaloids

- A large number of alkaloids are biosynthesized from tyrosine and they includes:
 - Simple phenylalkyl amines and tetrahydroisoquinolines
 - Colchicine
 - Emetine and related alkaloids
 - Tubocurarine
 - Opium alkaloids

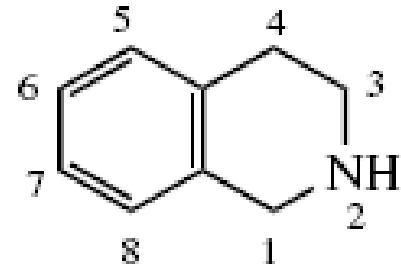
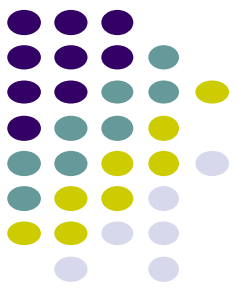


4.1. Simple phenylalkyl amines and tetrahydroisoquinolines

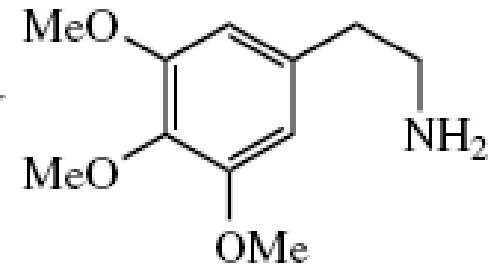


a) Mescaline (3, 4, 5-trimethoxy- β -phenethylamine)

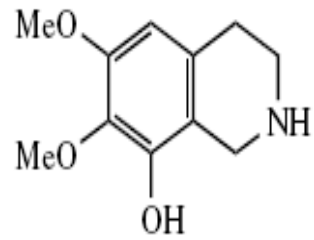
- it is the chief active constituent of the Mexican cactus, *Lophophora williamsii* or peyote
- is an alkaloid with psychoactive and hallucinogenic properties
 - ✓ A dose of 400mg causes vivid hallucination
 - ✓ It has been used to induce model psychosis
 - ✓ doesn't result in habituation and addiction
- Other constituents of peyote include anhalamine, anhalonidine, and anhalonine which are representatives of simple tetrahydroisoquinoline derivatives



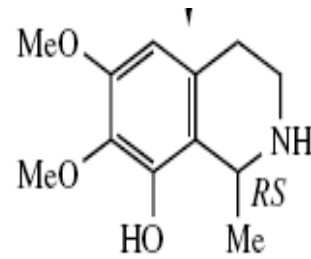
tetrahydroisoquinoline



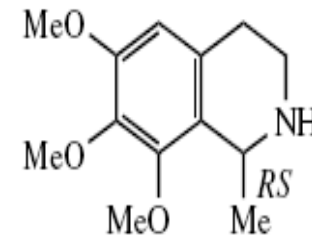
mescaline



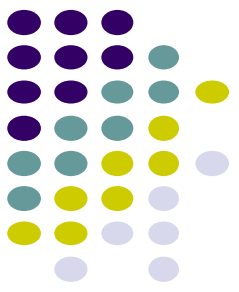
anhalamine



anhalonidine



anhalonine



b) Ephedra alkaloids

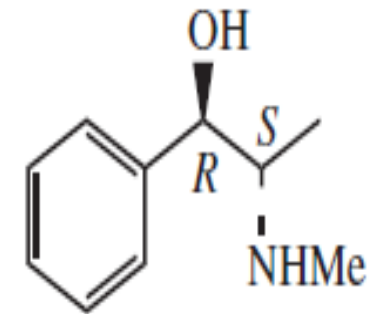
- Ephedra or Ma Huang is one of the oldest known drugs used by the Chinese for at least 5000 years.
- In Chinese characters “Ma” means astringent and “Huang” means yellow
- The plants typically contain 0.5–2.0% of alkaloids, according to species, and from 30–90% of the total alkaloids is (–)-ephedrine
- Related structures, including the diastereoisomeric (+)-pseudoephedrine and the demethyl analogues (–)-norephedrine and (+)-norpseudoephedrine are also present.
- In *E. intermedia*, the proportion of pseudoephedrine exceeds that of ephedrine.

Pharmacological actions

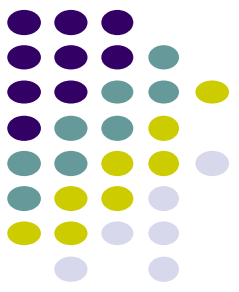
- Ephedrine is indirect sympathomimetic drug;
 - it causes increased blood pressure and pulse
 - contraction of blood vessels
 - dilation of the bronchi
- It also stimulates the CNS similar to amphetamines

Medicinal uses of Ephedrine

- treatment of bronchial asthma (allergies and colds)
- is used as a nasal decongestant in 1-3 % solution
- is a pressor agent and used to control hypotension in spinal anesthesia
- is used in narcolepsy
- treatment of hay fever



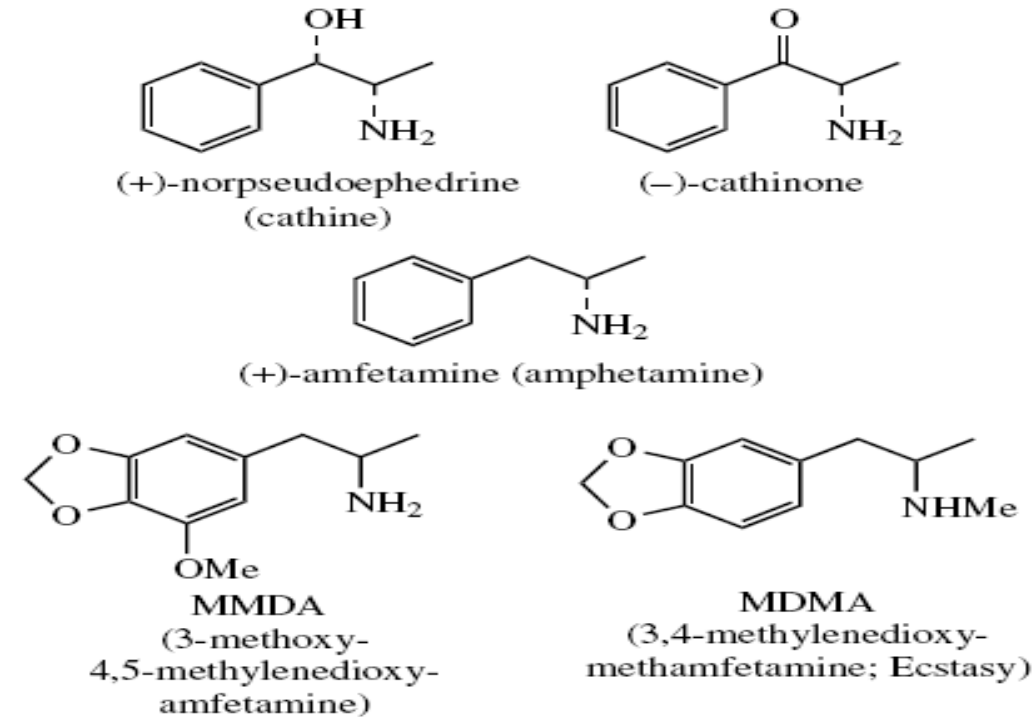
(-)-ephedrine

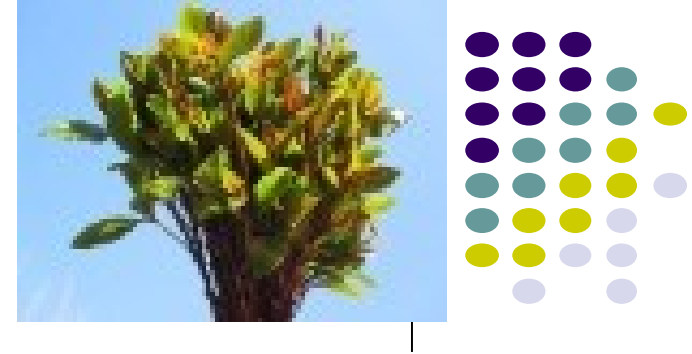


c) Khat alkaloids

- Khat, or Abyssinian tea, consists of the fresh leaves of *Catha edulis* (Celastraceae), a small tree cultivated in Ethiopia, East and South Africa, and the Yemen
- The plant is known under many names; it is called *qat* in Yemen, and *mira* in Kenya
- The leaves are widely employed in African and Arabian countries, where they **are chewed for a stimulant effect**
 - ✓ alleviates hunger and fatigue, but also gives a sensation of general well-being

- Lead to psychological dependence, but not normally physical dependence
- Dried leaves contain up to 1% cathine ((+)-norpseudoephedrine)
- Young fresh leaves contain (–)-cathinone as the **principal CNS stimulant**
- Cathinone has similar pharmacological properties as the synthetic CNS stimulant (amphetamine/dexamphetamine) with a similar potency
 - ✓ Both compounds act by inducing release of catecholamines
- Norephedrine and cathine do not contribute significantly to the psychostimulant action



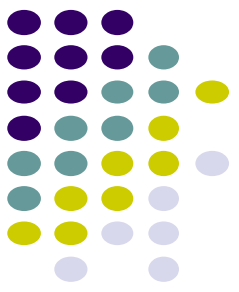


Desirable effects of Khat

- Relief from fatigue
- Increase alertness and energy level
- Enhance imaginative ability and capacity to associate ideas
- Feeling of elation
- Improve ability to communicate
- Heightened self confidence

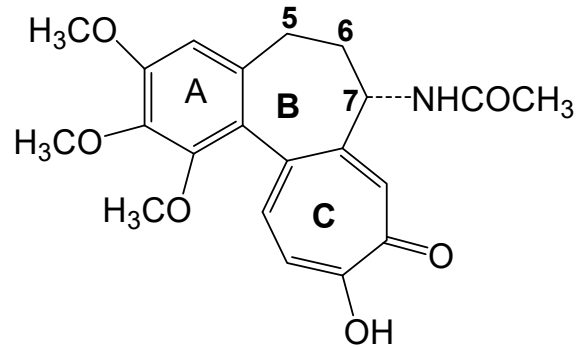
Undesirable effects of Chat

- Produces anorexia (loss of appetite) and constipation
- Brownish staining of the teeth
- Leads mental and nervous exhaustion
- Sexual dysfunction
- Produces insomnia
- Excessive consumption may lead to toxic psychosis
- Economic burden on the family



4.2. Colchicine

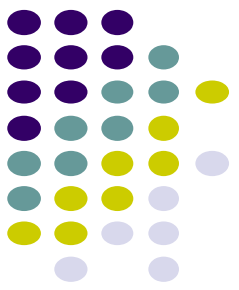
- obtained from *Colchicum autumnale* (Liliaceae)



- is an effective treatment for acute gouty attacks
- it is very toxic
- causes teratogenic birth defects

- not basic
- ring A and carbon 5, 6, and 7 are derived from phenylalanine
- tyrosine is the precursor of tropolone ring (ring C)

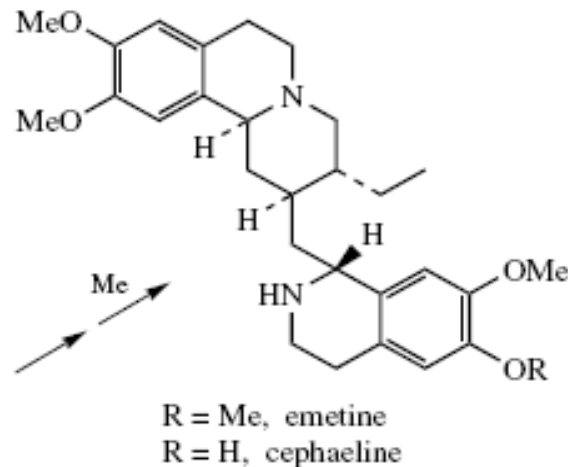




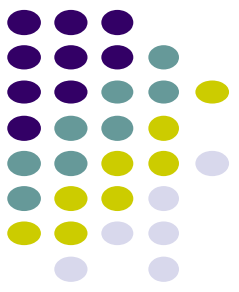
4.3. Emetine and related alkaloids

Biological sources:

- *Cephaelis ipecacuanha* (Rio or Brazilian ipecac) or *C. acuminata* (Cartagena, Nicaragua, or Panama ipecac) of the family Rubiaceae
- Ipecac contains 2–2.5% of alkaloids, the principal ones being **emetine** and **cephaeline**.

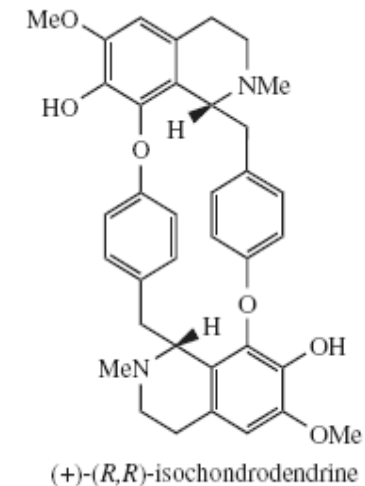
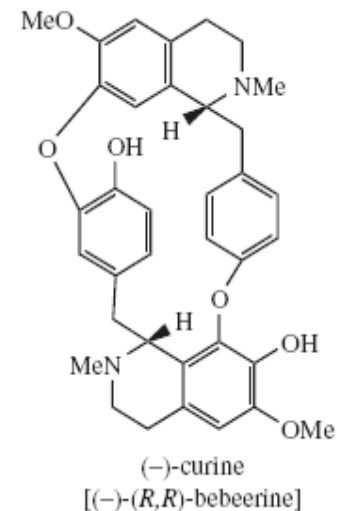
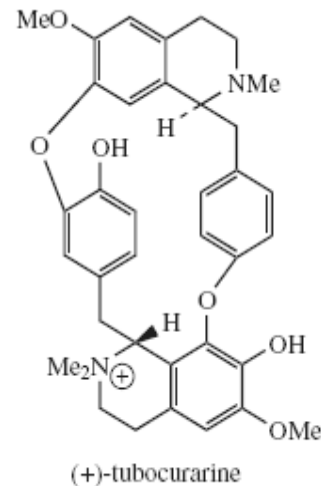


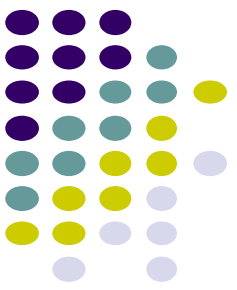
- emetine and the synthetic 2, 3-dehydroemetine have been useful as anti-amoebics
- used for drug overdose or poisoning
- has expectorant activity



4.4. Bisbenzylisoquinoline alkaloids: Tubocurarine

- Curare is the arrow poison of the South American Indians
 - ✓ represents the crude dried extract from the bark and stems of various plants
 - ✓ it may contain as many as 30 different plant ingredients
- the alkaloid content of curare is from 4% to 7%



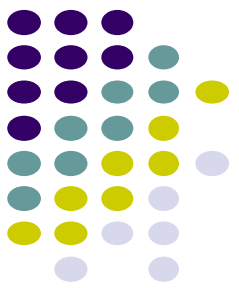


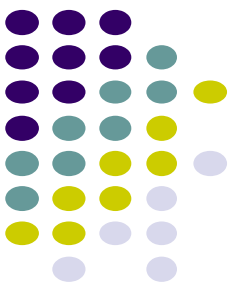
- kills by producing paralysis, a limp relaxation of voluntary muscles
 - ✓ competing with **acetylcholine** at **nicotinic receptor** sites, thus blocking nerve impulses at the neuromuscular junction.
 - ✓ thus found medicinal use as a muscle relaxant
- e.g. abdominal surgery, tonsillectomy, etc,
- where tense muscles needed to be relaxed
 - ✓ was also found to be of value in certain neurological conditions
- e.g. to temporarily relax rigid muscles and control **convulsions**, but was not a curative, multiple sclerosis, tetanus, and Parkinson's disease

4.5. Opium alkaloids

Opium

- is the air-dried milky exudate, or latex, obtained by incising the unripe capsules of the opium poppy *Papaver somniferum* (Papaveraceae)
- The term opium is from the Greek
 - ✓ *opion*, meaning **poppy juice**;
 - ✓ *Papaver* is the Latin name for the poppy;
 - ✓ *somniferum* is Latin meaning to produce sleep.



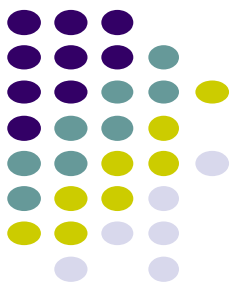


Ancient medical use

- Poppy was revered by several ancient societies for
 - ✓ its analgesic properties and its ability to induce sleep
- Opium latex has a long history of use for pain relieve and inducing sleep among Babylonia, Egyptian, Greek, and Roman civilizations
 - Opium eaten, drunk, and smoked
 - Most common method was to dissolve opium in alcohol – opium wine



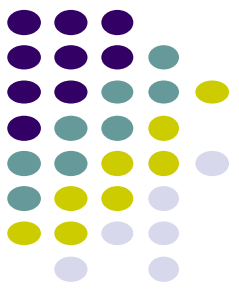
- Many alkaloids have been identified in opium latex (opiates) - maybe as **many as 50**
- The alkaloids are grouped into **four chemical classes**
 - Phenanthrenes (morphinans); Morphine, Codeine, Thebaine
 - Benzyloquinolines; Papaverine, noscapine (narcotine)
 - Phenylethylamines; Narceine
 - Diisoquinoline alkaloids
 - Only the phenanthrenes and Benzyloquinolines are used in medicine



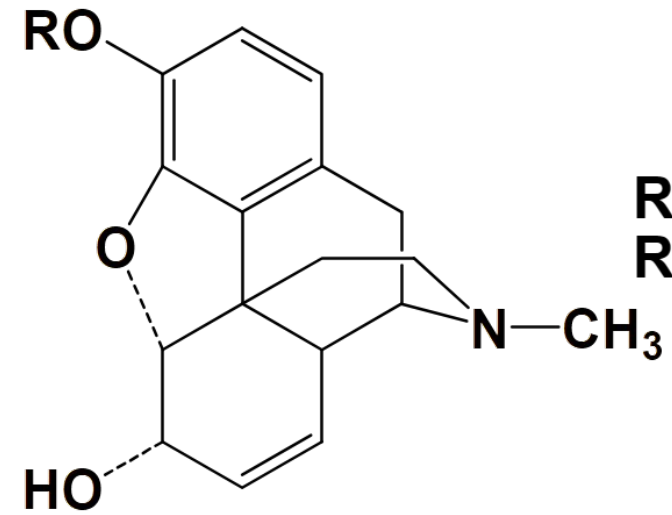
a) The phenanthrenes

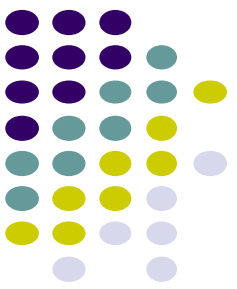
Morphine:

- first alkaloid identified from any plant in 1806 and the pain relieving properties soon recognized
- It constitute 4–21% of opium
- Morphine (like raw opium) is **strongly addictive** and because of this, it remains as one of the most valuable analgesics for relief of **severe pain**.
- The 4–ring system contained in morphine is called morphinan

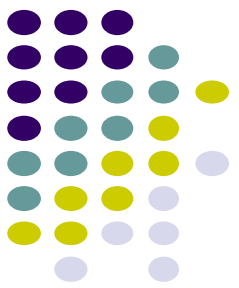


- Methylation of its phenolic hydroxyl group by using trimethyl phenyl ammonium sulphate gives **codeine**
- Addition of **ferric chloride** to neutral solution of morphine gives a **greenish blue colour**
- Morphine affects the central nervous system but also induces drowsiness and can depress respiration
 - ✓ overdose can cause death through respiratory failure



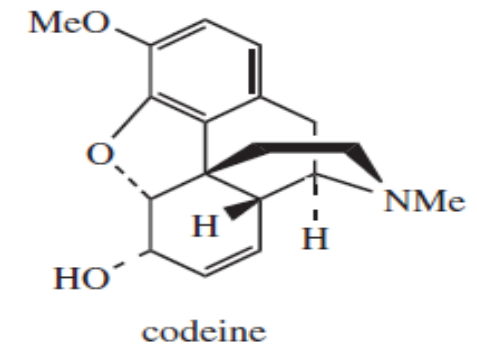


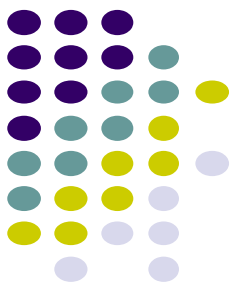
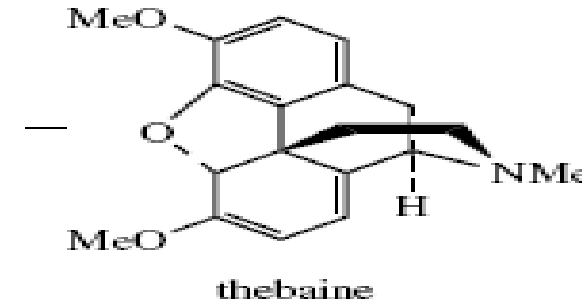
- Cause has high physical dependency
- The side-effect, constipation is utilized in some anti-diarrhoea preparations, e.g. kaolin and morphine
- Morphine is metabolized in the body to glucuronides, which are readily excreted
 - morphine 3-O-glucuronide is antagonistic to the analgesic effects of morphine
 - morphine 6-O-glucuronide is actually a more effective and longer lasting analgesic than morphine, with fewer side-effects



Codeine:

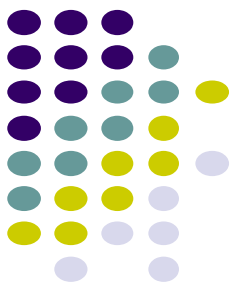
- 3-*O*-methyl ether of morphine and most widely used opiate
- Only 1/5 as strong as morphine and relatively safe non-addictive medium analgesic.
- also has valuable antitussive action (in cough syrups)
 - ✓ effectively depresses the cough centre or reflex
- relatively small amounts (0.2–0.7 %) found in opium, manufactured by semi-synthesis from morphine.





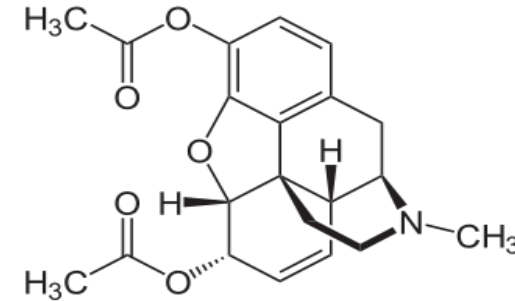
Thebaine:

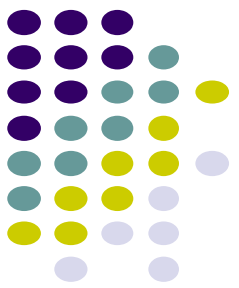
- Thebaine differs structurally from morphine/codeine mainly by its possession of a **conjugated diene ring system**
- It is almost devoid of analgesic and narcotic activity,
 - ✓ used as a morphine antagonist
- Its main value is as **substrate** for the semi-synthesis of other drugs such as the veterinary sedative, etorphine.
- It is the principal alkaloid in *Papaver bracteatum* which contains no morphine and codeine



Heroin:

- Heroin is a semi-synthetic derivative of morphine - diacetyl morphine
- In 1898, Bayer Pharmaceuticals introduced heroin which they thought was a non-addictive opiate and more effective than morphine and codeine
- Within a few years over 1 million people addicted to heroin in US alone
 - ✓ Soon realized that it was actually more addictive than morphine - **actually six times more addictive**
 - ✓ In US use of heroin is illegal placed under Harrison Act of 1914
- Heroin still used medicinally in other countries
 - ✓ in cough syrups
- Heroin addiction a major problem worldwide

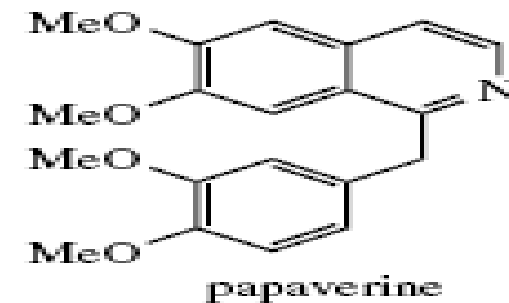




b) Benzyloquinolines

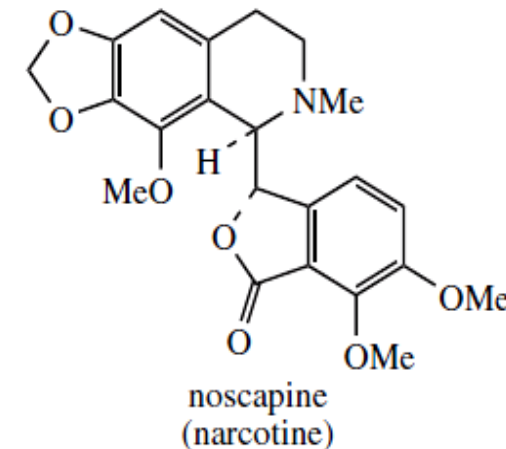
Papaverine:

- **one of the simplest** opium alkaloids that can be isolated from opium (1%)
- It has little or no analgesic or hypnotic properties
 - ✓ possesses spasmolytic, vasodilator and smooth muscle relaxant activity
 - ✓ via its Ca^{2+} channel blocking ability.



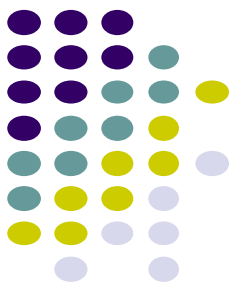
Noscapine

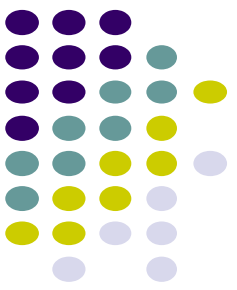
- Noscapine is a member of the phthalideisoquinoline alkaloids
- has good antitussive and cough suppressant activity comparable to that of codeine
 - ✓ no analgesic or narcotic action
 - ✓ may have teratogenic properties
- In recent studies, antitumour activity has been noted from noscapine



Papaveretum

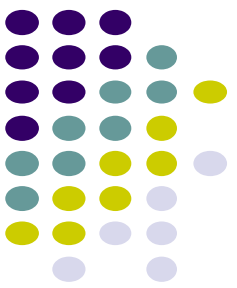
- is a mixture of purified opium alkaloids, as their hydrochlorides
- is now formulated to contain only
 - morphine HCl (85.5%)
 - codeine HCl (7.8%)
 - papaverine HCl (6.7%)
- It is used for pain relief during and after operations





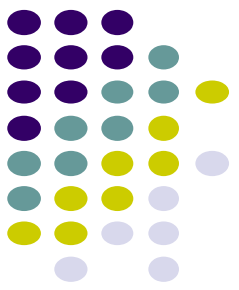
Opioids

- A vast range of semi-synthetic or totally synthetic morphine-like derivatives
- Many have similar narcotic and pain-relieving properties as morphine, but are **less habit forming**
- Others possess the cough-relieving activity of codeine without the analgesic effect
- More than 90% of the morphine extracted from opium (or poppy straw) is currently processed to give other derivatives



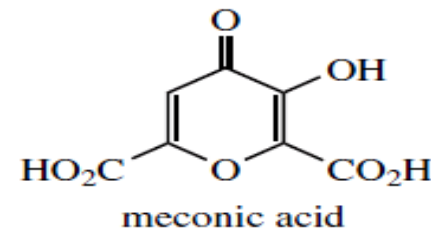
Examples:

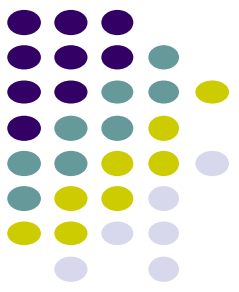
- Pethidine (meperidine) is one of the most widely used synthetic opiates
 - ✓ less potent than morphine, but produces prompt, short-acting analgesia,
 - ✓ is also less constipating than morphine. It can be addictive
- Tramadol is a recent drug claimed to produce analgesia by two mechanisms,
 - ✓ an opioid mechanism
 - ✓ also by enhancement of serotonergic and adrenergic pathways.



Detection of Opium Alkaloids

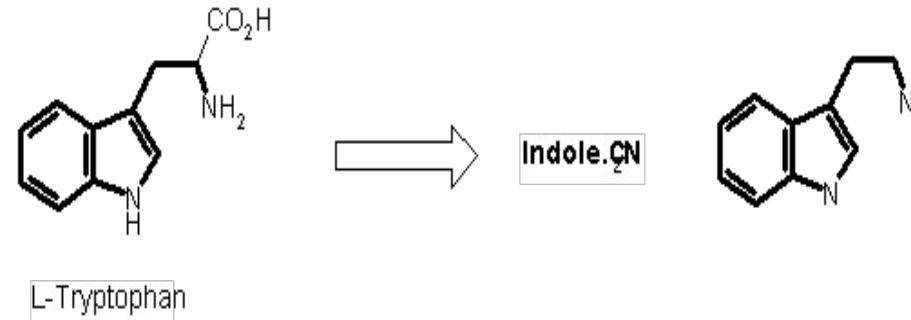
- Opium alkaloids usually occur in salt form combined with a unique acid known as **meconic acid**
 - ✓ it occurs invariably in Papaver species and has not been detected in other plant species
 - ✓ it gives a deep red-coloured complex with ferric chloride
 - ✓ and thus presence of meconic acid can be used to detect the presence of opium
- This has been used as a rapid and reasonably specific test for opium



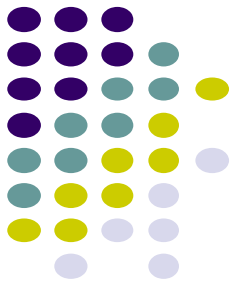


5. Tryptophan Derived Alkaloids

- Decarboxylation of tryptophan often accompanied with hydroxylation and subsequent reactions with
 - acetate (C-2)
 - pyruvate (C-3)
 - mevalonate (C-5)
 - irridoid (C-10) units
- give rise to a wide series of terpenoid indole Alkaloids
- More than 3000 terpenoid indole alkaloids are recognized
- They are found mainly in eight plant families, of which the Apocynaceae, the Loganiaceae, and the Rubiaceae provide the best sources

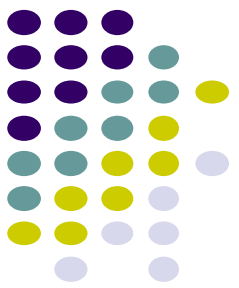


Tryptophan Derived Alkaloids



Terpenoid indole alkaloids include

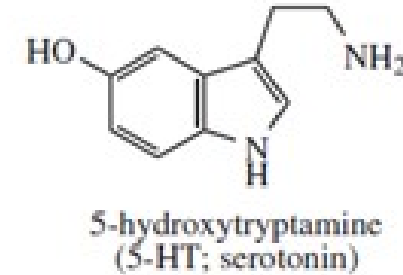
- Simple indoles
- Tricyclic alkaloids
- Ergot alkaloids
- Rauwolfia alkaloids
- Strychnos alkaloids
- Cinchona alkaloids
- Perwinkle alkaloids

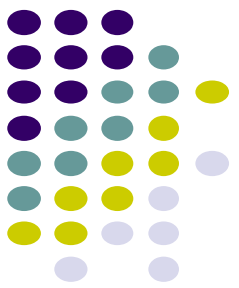


1.Simple indoles

A) 5-Hydroxytryptamine (Serotonin)

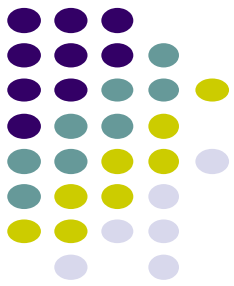
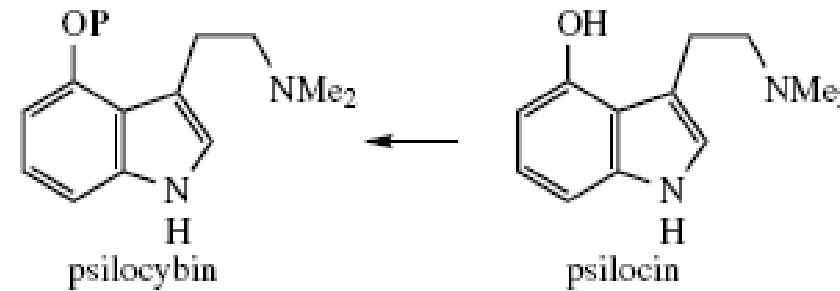
- is a monoamine neurotransmitter
- found in:
 - cardiovascular tissue
 - the peripheral nervous system,
 - blood cells
 - the central nervous system
- It also occurs in stinging nettles *Urtica* spp, in banana, *Musa* spp,
- It mediates many central and peripheral physiological functions:
 - contraction of smooth muscle
 - vasoconstriction,
 - food intake, pain perception, memory, sleep





- Platelets and neurons possess a high affinity 5-HT reuptake mechanism
- This mechanism may be inhibited by widely-prescribed antidepressant drugs.
 - selective serotonin re-uptake inhibitors (SSRIS),
e.g. fluoxetine (Prozac)
 - thereby increasing levels of 5-HT in the CNS
- Migraine headaches that do not respond to analgesics may be relieved by the use of an agonist of the 5-HT receptor
 - ✓ Example sumatriptan, naratriptan, rizatriptan, and zolmitriptan

Psilocybin and psilocin



- the genus *Psilocybe* ‘magic mushrooms’ constitutes a group of small mushrooms with worldwide distribution
- over 80 species of *Psilocybe* have been found to be psychoactive
- *Psilocybe mexicana* has been used by the Mexican Indians in ancient ceremonies for many years
- ingestion of the fungus causes visual hallucinations:
 - ✓ rapidly changing shapes and colors
 - ✓ different perceptions of space and time,
- the effects gradually wearing off and causing no lasting damage or addiction
- the active hallucinogens, present at about 0.3%, are the **tryptamine derivatives psilocybin and psilocin**
 - ✓ are structurally related to the neurotransmitter 5-HT

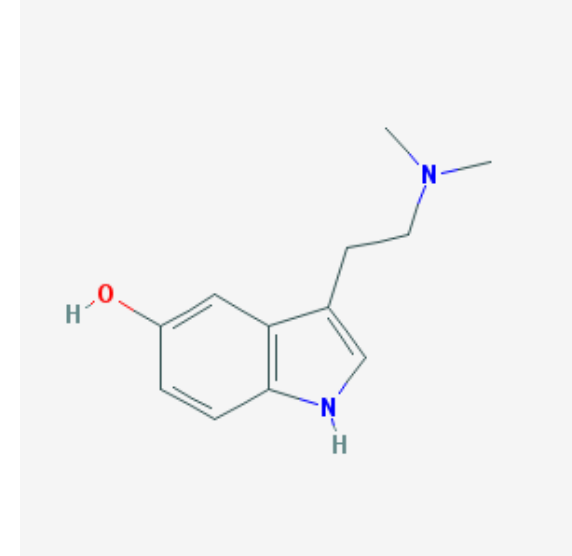


- induce model psychoses
- psilocybin is chemically more stable than psilocin.

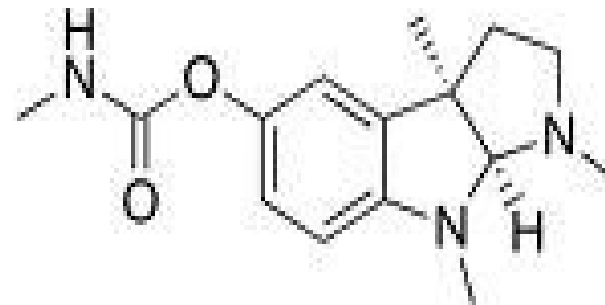


Bufotenin

- obtained from the parotid glands of the toad *Bufo vulgaris*
- it also occurs in a **number of flowering plants** including *Piptadenia* spp
- ingredients of some South American narcotic snuff



2. Tricyclic alkaloids



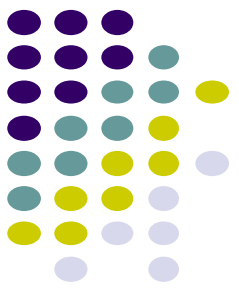
Physostigma alkaloids

- *Physostigma venenosum* (Fabaceae) is a perennial woody climbing plant found on the banks of streams in West Africa
- The seeds are known as Calabar beans and have an interesting history in the native culture as an ordeal poison
- The seeds contain several alkaloids, the major one being physostigmine (eserine)
- The unusual pyrroloindole ring system is also present in some of the minor alkaloids, e.g. eseramine
- Solutions of physostigmine are not particularly stable in the presence of air and light, especially under alkaline conditions, oxidizing to a red quinone, rubeserine

Pharmacologic action

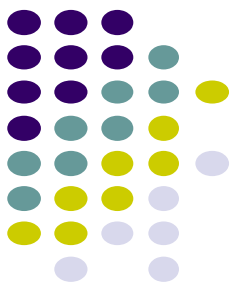
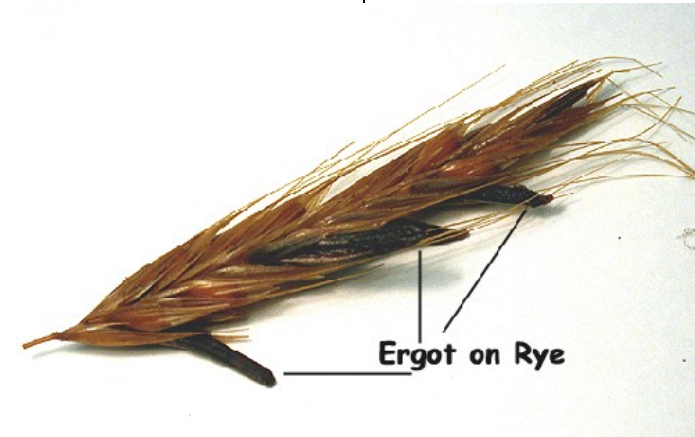
Physostigmine (eserine)

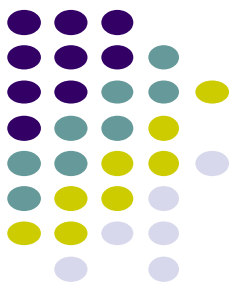
- is a reversible inhibitor of cholinesterase and hence prevent normal destruction of acetylcholine
 - antidote to anticholinergic poisons such as hyoscyamine/atropine
- it also reverses the effects of competitive muscle relaxants such as tubocurarine, atracurium, etc
- Its major use is as a miotic often to combat the effect of mydriatics such as atropine
- It is a valuable treatment for glaucoma, often in combination with pilocarpine
- The biological activity resides primarily in the carbamate portion



3. Ergot alkaloids or ergolines

- Medicinal ergot is the dried sclerotium of the fungus *Claviceps purpurea* (Clavicipitaceae) developed on the ovary of rye
- Ergots may be harvested with the grain and contaminate flour or animal feed
- The consumption of ergot-infected rye has resulted in the disease ergotism
- There are three broad clinical features of ergot poisoning
 - Alimentary upsets, e.g. diarrhoea, abdominal pains, and vomiting
 - Circulatory changes, e.g. coldness of hands and feet due to a vasoconstrictor effect, a decrease in the diameter of blood vessels, especially those supplying the extremities
 - Neurological symptoms, e.g. headache, vertigo, convulsions, psychotic disturbances, and hallucinations.

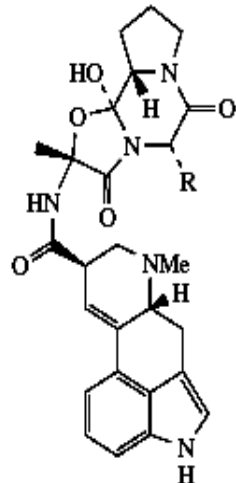
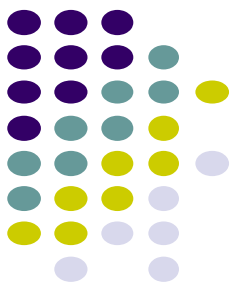




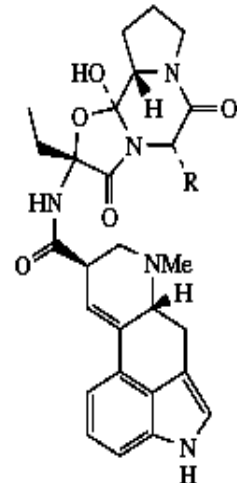
- The medicinally useful compounds are derivatives of (+)-lysergic acid, and can be classified into two groups
 - the water-soluble amino alcohol derivatives (up to 20% of the total alkaloids)

E.g. Ergometrine is an amide of lysergic acid and 2-aminopropanol
 - water-insoluble peptide derivatives (up to 80% of the total alkaloids)
 - contain a cyclized tripeptide fragment bonded to lysergic acid via an amide linkage
 - Based on the nature of the three amino acids, these structures can be subdivided into three groups
 - the ergotamine group,
 - the ergoxine group, and
 - the ergotoxine group

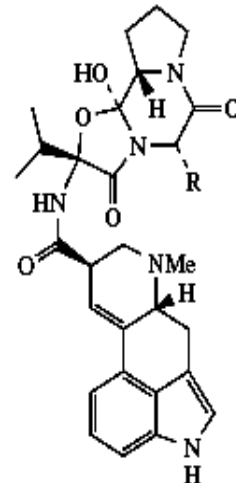
- The amino acids involved are **alanine**, valine, leucine, isoleucine, **phenylalanine**, **proline**, and α -aminobutyric acid, in various combinations
- All contain proline in the tripeptide



ergotamine group



ergoxine group



ergotoxine group

R = CH₂Ph

ergotamine

R = CH₂CHMe₂

ergosine

R = CH(Me)Et
S

[β -ergosine]

R = CHMe₂

ergovaline

R = Et

ergobine

ergostine

ergoptine

[β -ergoptine]

ergonine

ergobutine

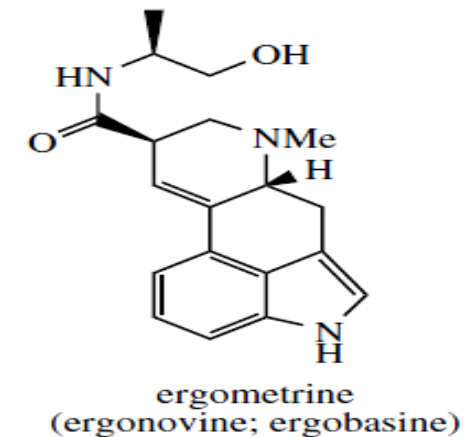
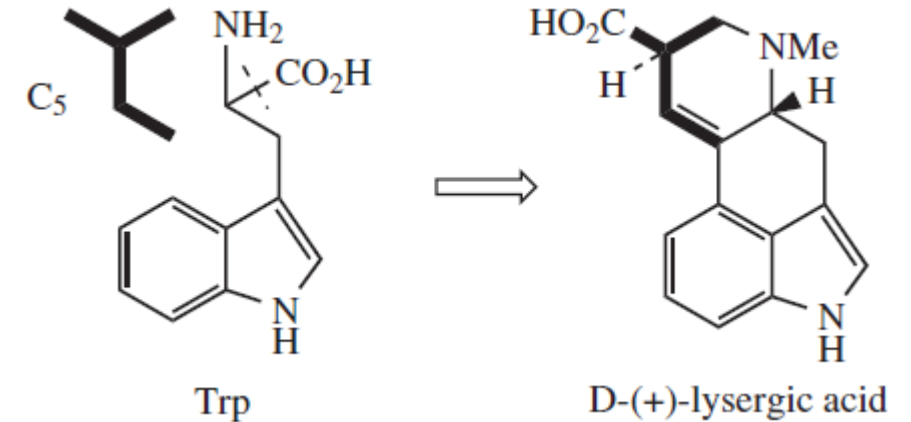
ergocristine

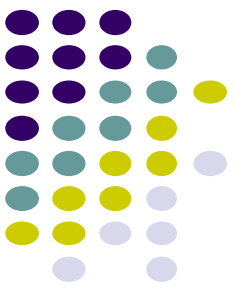
α -ergocryptine

β -ergocryptine

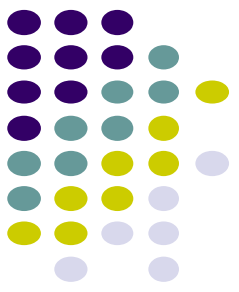
ergocornine

ergobutyryne

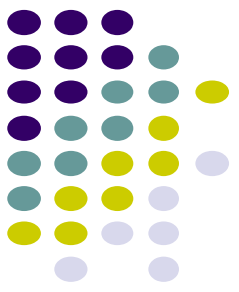




- The ergot alkaloids act at α -adrenergic, dopaminergic and serotonergic receptors
- Ergometrine (ergonovine) is used as an oxytocic
 - injected during the final stages of labour
 - immediately following childbirth, especially if haemorrhage occurs
- It is used in the treatment of acute attacks of migraine
- Ergotamine
 - produces a pronounced peripheral vasoconstrictor action.
 - ✓ acute attacks of migraine
 - ✓ combined with caffeine, e.g. Cafergot[®],



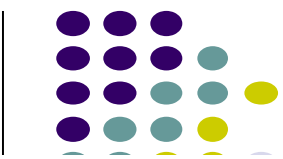
- A number of semi-synthetic lysergic acid derivatives **act by stimulation of dopamine receptors in the brain**, and are of value in the treatment of neurological disorders such as Parkinson's disease
 - ✓ Bromocriptine (2-bromo- α -ergocryptine), cabergoline, lisuride (lysuride), and pergolide are all used in this way
 - ✓ Bromocriptine and cabergoline find wider use in that they also inhibit release of prolactin and used in the treatment of **breast tumours**
 - ✓ Methysergide is a semi-synthetic analogue of ergometrine,
 - ✓ is a potent 5-HT antagonist and prophylaxis of severe migraine headaches, though its administration has to be very closely supervised



4. Rauwolfia alkaloids

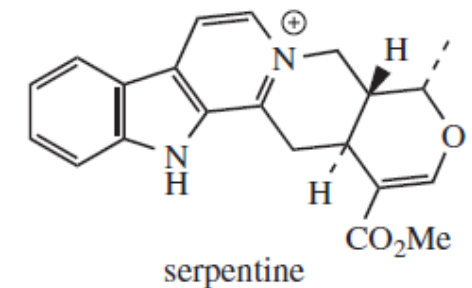
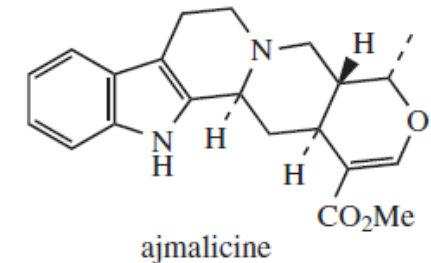
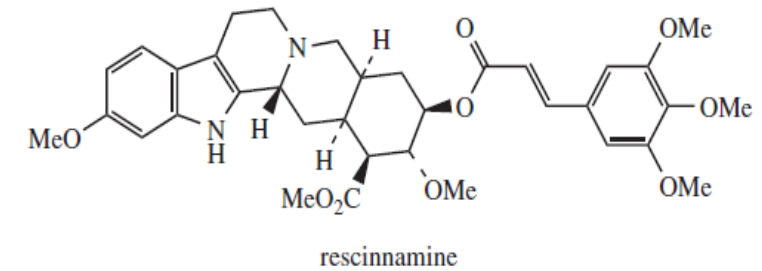
- Rauwolfia is the dried rhizome and roots of *Rauwolfia* (sometimes *Rauvolfia*) *serpentina* (Apocynaceae) or snakeroot,
 - a small shrub from India, Pakistan, Burma, and Thailand
- Other species used in commerce include *R. vomitoria* from tropical Africa and *R. canescens* (= *R. tetraphylla*) from India and the Caribbean
- Rauwolfia has been used in Africa for hundreds of years, and in India for at least 3000 years
- It was used as :
 - an antidote to snake-bite,
 - to remove white spots in the eyes,
 - against stomach pains,
 - fever, vomiting, and headache, and
 - to treat insanity





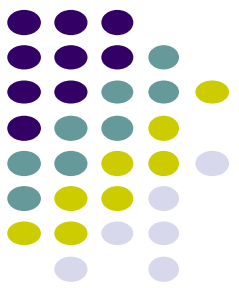
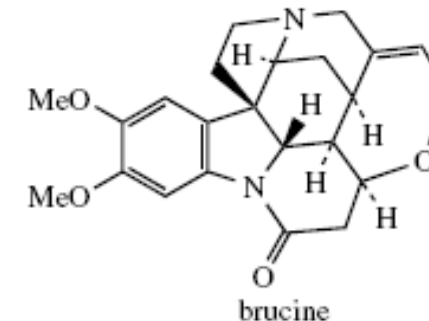
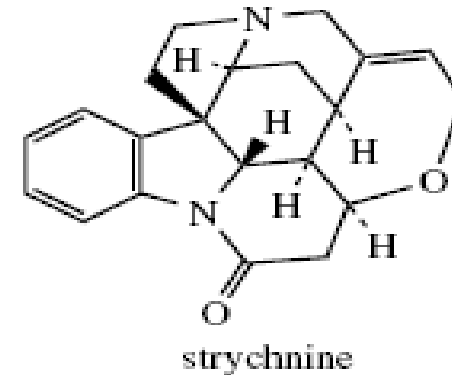
Chemical constituents

- *Rauwolfia serpentina* contains a wide range of indole alkaloids
 - E.g. reserpine, rescinnamine, deserpidine, ajmalicine and ajmaline
- **Reserpine and deserpidine** are major alkaloids in *R. canescens*
 - ✓ Reserpine and deserpidine have been widely used as antihypertensives and mild tranquillizers
 - They act by interfering with catecholamine storage
 - depleting levels of available neurotransmitters
- Both ajmalicine (= raubasine) and ajmaline are used clinically in Europe
 - ajmalicine is employed as an antihypertensive
 - ajmaline is of value in the treatment of cardiac arrhythmias



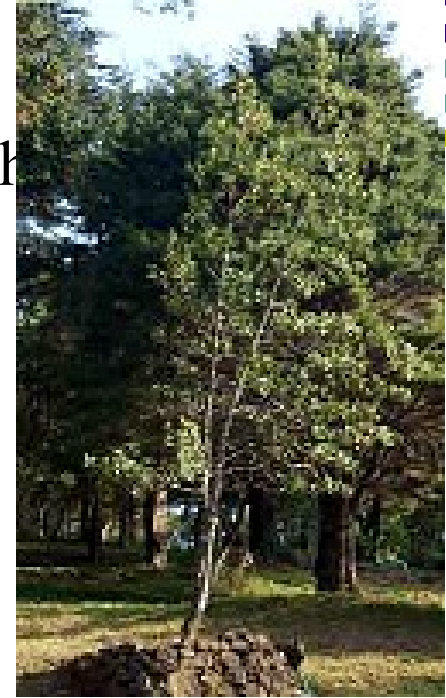
5. Strychnos alkaloids

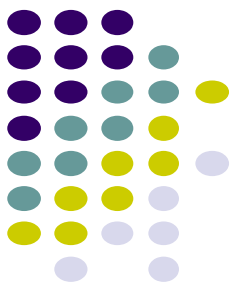
- Nux-vomica consists of the dried ripe seeds of *Strychnos nux-vomica* (Loganiaceae), a small tree found in a wide area of East Asia
 - ✓ contain 1.5–5% of alkaloids, chiefly strychnine (about 1.2%) and brucine
- Strychnine is very toxic, affecting the CNS and causing convulsions
 - ✓ Fatal poisoning (consumption of about 100 mg by an adult) would lead to asphyxia following contraction of the diaphragm
 - ✓ Its only medicinal use is in very small doses as an appetite stimulant and general tonic
- Brucine is considerably less toxic
- Seeds of *Strychnos ignatii* have also served as a commercial source of strychnine and brucine



6. *Cinchona* alkaloids

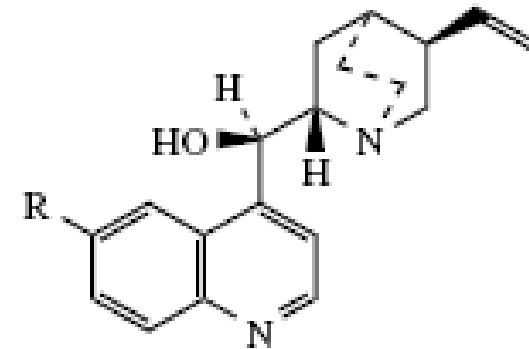
- *Cinchona* bark is the dried bark from the stem and root of species of *Cinchona* (Rubiaceae), which are large trees indigenous to South America
- About a dozen different *Cinchona* species have been used as commercial sources:
 - *C. succirubra* provides 'red' bark (alkaloid content 5–7%),
 - *C. ledgeriana* gives 'brown' bark (alkaloid content 5–14%)
 - *C. calisaya* 'yellow' bark with an alkaloid content of 4–7%
- Selected hybrids can yield up to 17% total alkaloids
- Bark is stripped from trees which are about 8–12 years old
- The major alkaloids are quinine, quinidine, cinchonidine, and cinchonine
- The alkaloids are often present in the bark in salt combination with **quinic acid** or a tannin material called **cinchotannic acid**





Quinine

- treatment of multidrug-resistant malaria
- Is consumed in beverages (tonic water)
- Quinine also is a skeletal muscle relaxant
 - finds use in the prevention and treatment of nocturnal cramp

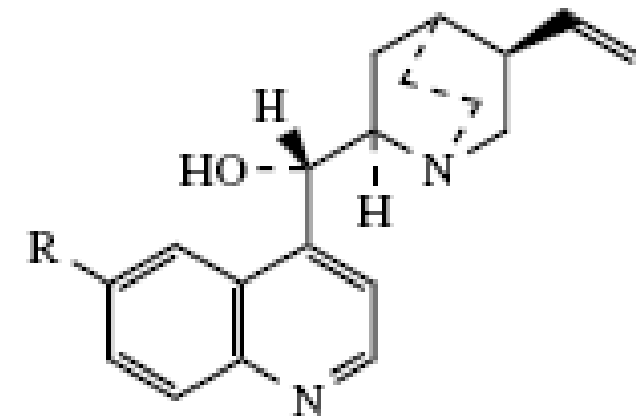


R = H, cinchonidine

R = OMe, quinine

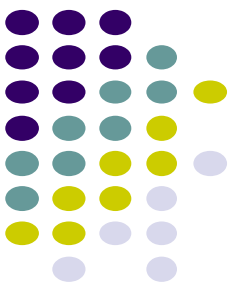
Quinidine

- is administered to treat cardiac arrhythmias



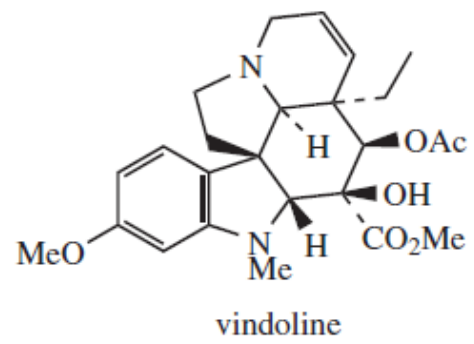
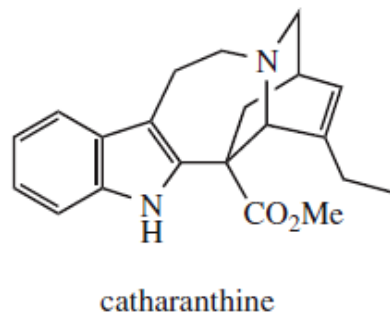
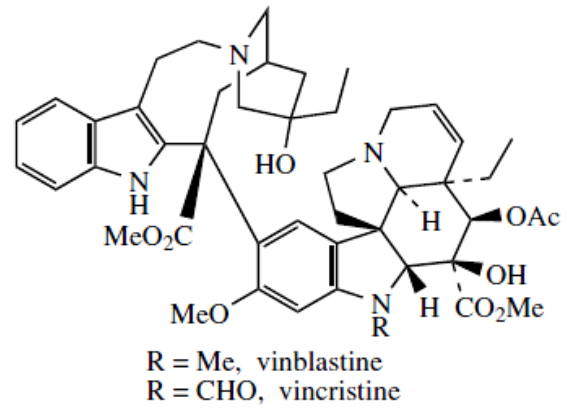
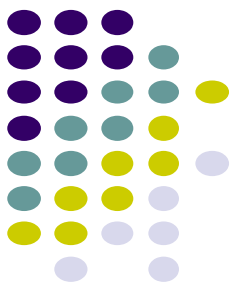
R = H, cinchonine

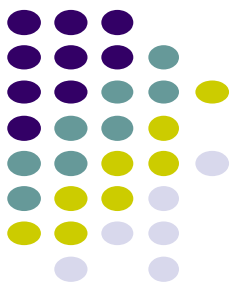
R = OMe, quinidine



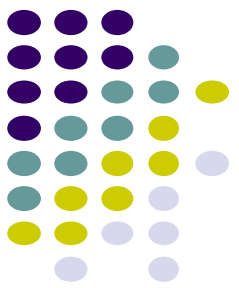
7. Periwinkle alkaloids

- The Madagascar periwinkle *Catharanthus roseus* (= *Vinca rosea*) (Apocynaceae) is a small herb or shrub originating in Madagascar.
- More than 150 alkaloids have been characterized in the plant
- The most important ones are dimeric indole alkaloid structures (more correctly bis-indole alkaloids):
 - Vinblastine vinleurosine
 - vinrosidine vincristine
- The alkaloids vinblastine and vincristine are extremely valuable drugs in cancer chemotherapy
- Despite the minor difference in structure between vinblastine and vincristine, a significant difference exists in the spectrum of human cancers that respond to the drugs.

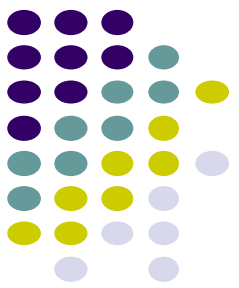




- only very small amounts of these desirable alkaloids are present in the plant
 - over 500 kg of catharanthus is needed to yield 1 g of vincristine (0.0002% yield)
 - Extraction is both costly and tedious
- The growing importance of vincristine relative to vinblastine as drugs is not reflected in the plant,
 - produces a much higher proportion of vinblastine
- It is possible to convert vinblastine into vincristine by:
 - controlled chromic acid oxidation
 - via a microbiological N-demethylation using *Streptomyces albogriseolus*
- Ajmalicine is present in the roots of *C. roseus* at a level of about 0.4%
 - this plant is used as a commercial source in addition to *R. serpentina*

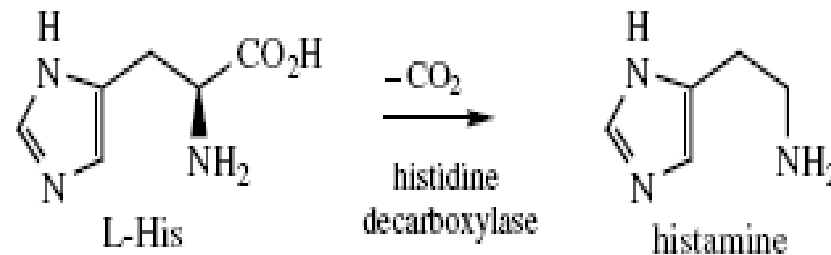


- Vinblastine
 - Hodgkin's disease, a cancer affecting the lymph glands, spleen, and liver
- Vincristine
 - has superior antitumour activity compared to vinblastine
 - is more neurotoxic
 - It is clinically more important than vinblastine,
 - useful in the treatment of childhood leukaemia
 - lymphomas, small cell lung cancer, cervical and breast cancers, also respond favorably
- The alkaloids need to be injected,
- Both generally form part of a combination with anticancer drugs
- Vindesine is a semi-synthetic derivative of vinblastine,
 - for the treatment of acute lymphoid leukaemia in children
- Vinorelbine, an anhydro derivative of 8-norvinblastine,
 - It is orally active
 - has a broader anticancer activity
 - with lower neurotoxic side-effects



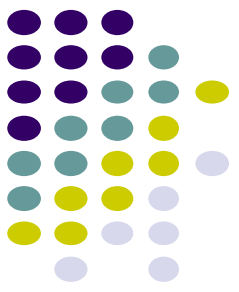
6. Histidine Derived Alkaloids

- The amino acid L-histidine contains an **imidazole ring**
 - is thus the likely precursor of alkaloids containing this ring system

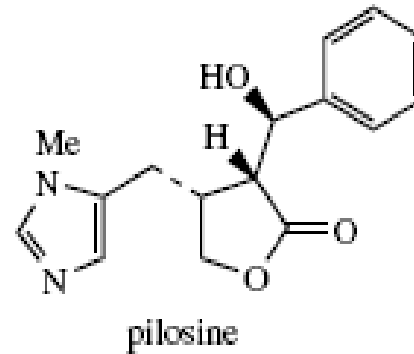
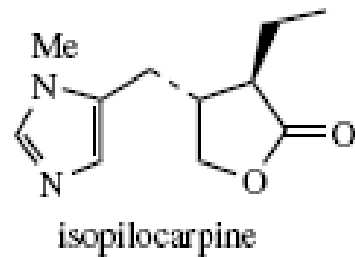
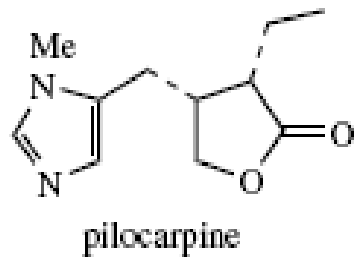


Pilocarpine

- Pilocarpus or jaborandi consists of the dried leaflets of *Pilocarpus jaborandi*, *P. microphyllus*, or *P. pennatifolius* (Rutaceae)
 - small shrubs from Brazil and Paraguay
- *Pilocarpus microphyllus* is currently the main source
 - ✓ The alkaloid content (0.5–1.0%)
 - ✓ Consists **principally of the imidazole alkaloid pilocarpine**, together with small amounts of pilosine and related structures



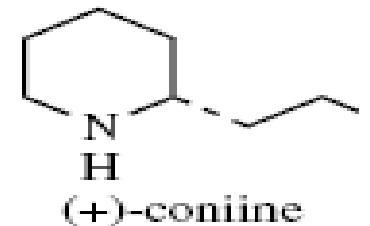
- The *iso* alkaloids lack biological activity
- The alkaloid content of the leaf rapidly deteriorates on storage
- Pilocarpine is a cholinergic agent and stimulates the muscarinic receptors in the eye.
 - constriction of the pupil
 - enhancement of outflow of aqueous humour



7. Acetate-derived Alkaloids

Coniine

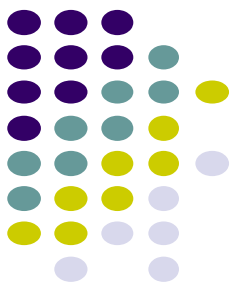
- *Conium maculatum* (Umbelliferae/Apiaceae) or poison hemlock
- This plant can be differentiated from most other members of the Umbelliferae/Apiaceae by its smooth purple-spotted stem
- The ancient Greeks are said to have executed condemned prisoners, including Socrates, using poison hemlock
 - ✓ The poison causes gradual muscular paralysis followed by convulsions and death from respiratory paralysis
- All parts of the plant are poisonous due to the alkaloid content
 - though the highest concentration of alkaloids is found in the green fruit (up to 1.6%)
- The major alkaloid (about 90%) is the volatile liquid coniine
 - smaller amounts of structurally related piperidine alkaloids: N-methylconiine and γ -coniceine are present



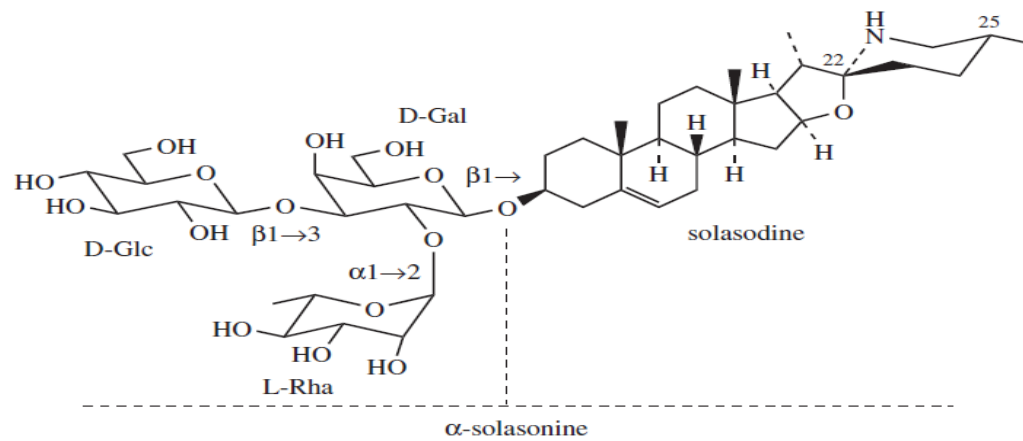
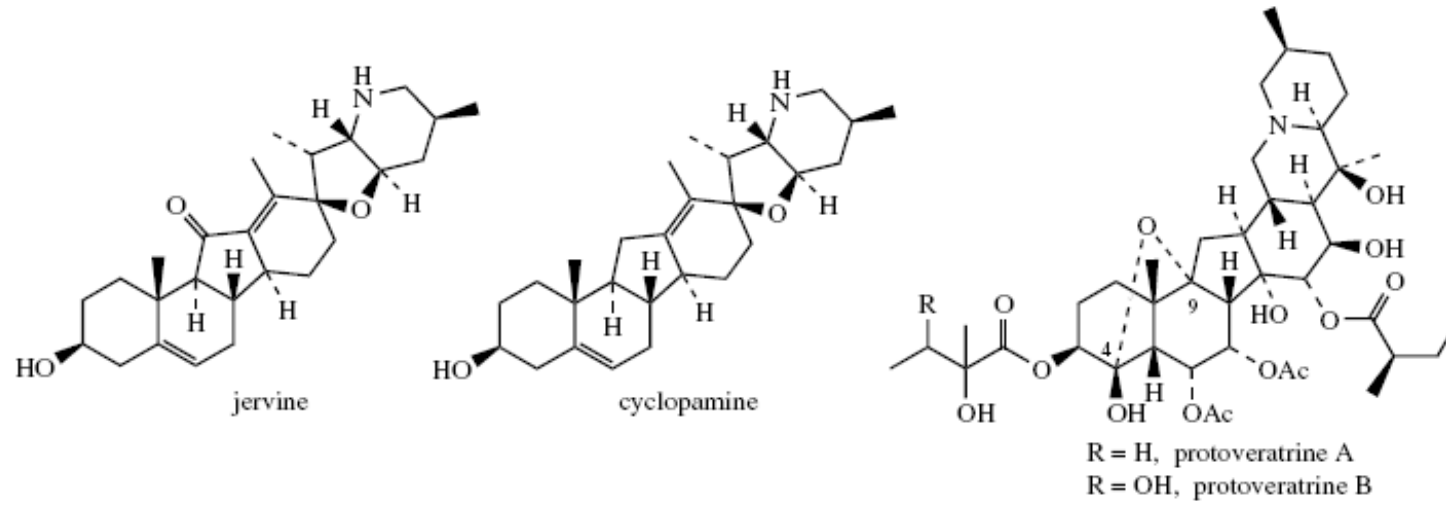
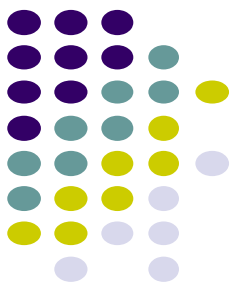
7. Steroidal Alkaloids

- Many plants in the Solanaceae accumulate steroidal alkaloids based on a C_{27} cholestane skeleton
 - e.g. solasodine found in potatoes (*Solanum tuberosum*)
 - tomatidine from tomato (*Lycopersicon esculente*)
- These are essentially nitrogen analogues of steroidal saponins
 - have surface activity
 - haemolytic properties
- They are usually present as glycosides
- Several plants in the Liliaceae, notably the genus *Veratrum* (Liliaceae/Melanthiaceae), contain group of steroidal alkaloids.
 - a fundamental change to the basic steroid nucleus has taken place

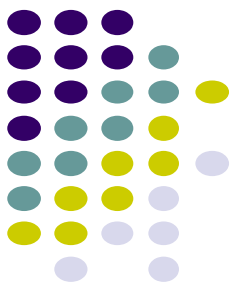
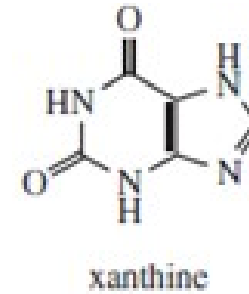




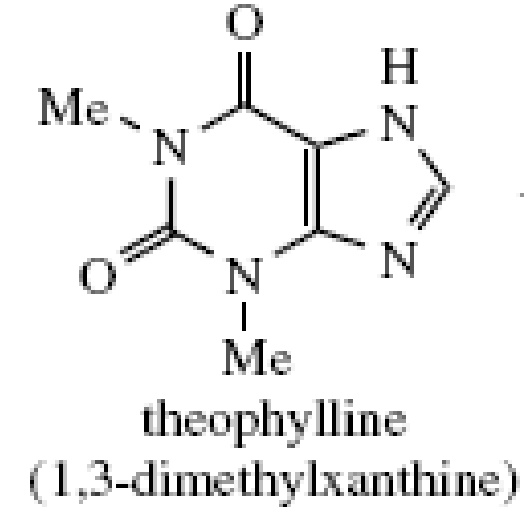
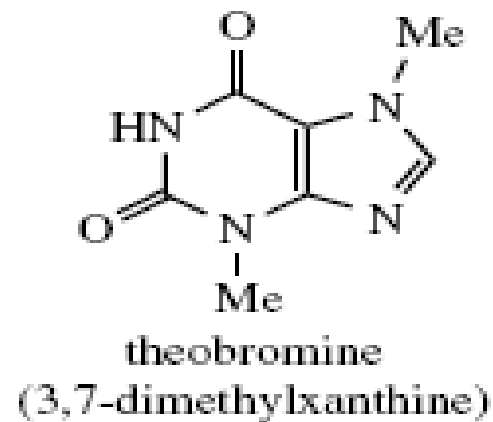
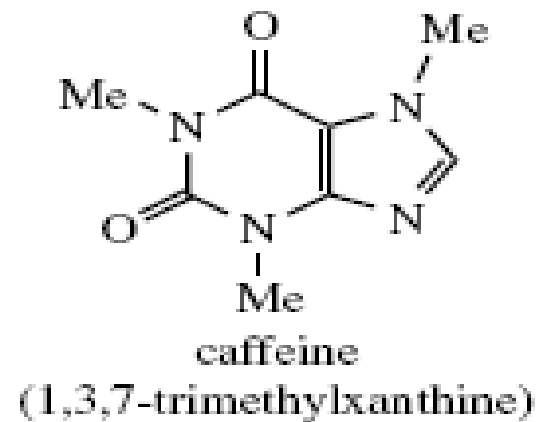
- Jervine and cyclopamine from *Veratrum californicum* are toxic components in this plant that are responsible for severe teratogenic effects
 - ✓ Animals grazing on *V. californicum* and some other species of *Veratrum* frequently give birth to young with cyclopia
 - a malformation characterized by a single eye in the centre of the forehead
- Other *Veratrum* alkaloids, especially those found in *V. album* and *V. viride*, have been employed medicinally as hypotensive agents, These medicinal alkaloids
 - e.g. protoveratrine A and protoveratrine B

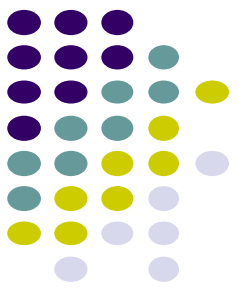


8. Purine Alkaloids

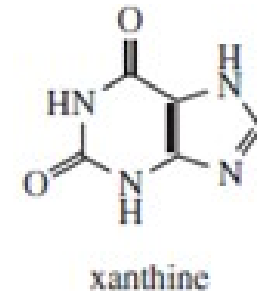


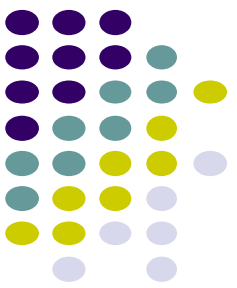
- Caffeine, theobromine, and theophylline are usually referred to as purine alkaloids



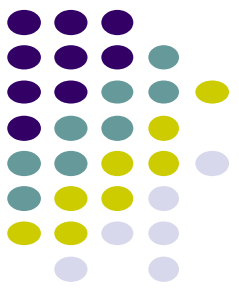


- The purine alkaloids are all methyl derivatives of xanthine
 - ✓ they commonly co-occur in a particular plant
 - ✓ they are water-soluble
- They competitively inhibit phosphodiesterase.
 - resulting in an increase in cyclic AMP
 - subsequent release of adrenaline
 - **a stimulation of the CNS,**
 - **a relaxation of bronchial smooth muscle,**
 - **induction of diuresis, as major effects**
 - tea, coffee, cocoa, and cola





- These effects vary in the three compounds
 - Caffeine
 - the best CNS stimulant
 - has weak diuretic action
 - Theobromine
 - little stimulant action
 - has more diuretic activity
 - muscle relaxant properties
 - Theophylline
 - low stimulant action
 - is an effective diuretic,
 - relaxes smooth muscle better than caffeine or theobromine
 - **aminophylline**
 - **choline theophyllinate**



Coffee

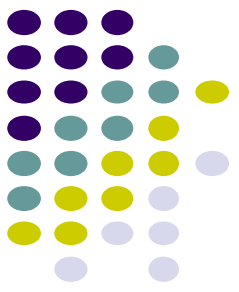
- consists of the dried ripe seed of *Coffea arabica*, *C. canephora*, *C. liberica*, or other *Coffea* species (Rubiaceae)
- seeds contain 1–2% of caffeine and traces of theophylline and theobromine
 - ✓ They are mainly combined in the green seed with **chlorogenic acid** (5–7%)
 - ✓ roasting releases them from their salt and also causes some decomposition of chlorogenic acid to quinic acid and caffeic acid
- The nicotinic acid derivative trigonelline is present in green seeds to the extent of about 0.25–1%
 - during roasting, this is extensively converted into nicotinic acid (vitamin B₃)
- Volatile oils and tannins provide odour and flavour



Tea

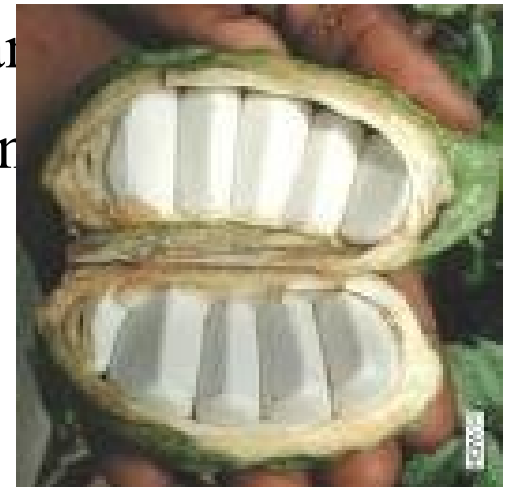
- Tea is the prepared leaves and leaf buds of *Camellia sinensis* (*Thea sinensis*) (Theaceae)
- **black tea**, the leaves are allowed to ferment, allowing enzymic oxidation of polyphenols
 - During oxidation, colourless catechins (up to 40% in dried leaf) are converted into intensely coloured **theaflavins** and **thearubigins**
 - ✓ Theaflavins are believed to act as **radical scavengers/antioxidants**, and to provide beneficial effects against cardiovascular disease, cancers, and the ageing process generally
- **green tea** is produced by steaming and drying the leaves to prevent oxidation
- **Oolong tea** is semi-fermented, drying under the strong sun and oxidation before curling and twisting
- contains 1–4% caffeine,
- small amounts (up to 0.05%) of both theophylline and theobromine
- Astringency and flavour come from tannins and volatile oils.
 - the latter containing monoterpene alcohols (geraniol, linalool) and aromatic alcohols (benzyl alcohol, 2-phenylethanol)

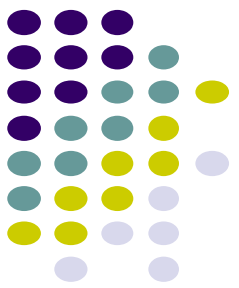




Cola

- Cola, or kola, is the dried cotyledon from seeds of various species of Cola (Sterculiaceae)
e.g. *C. nitida* and *C. acuminata*
- Cola seeds contain up to
 - 3% caffeine
 - 0.1% theobromine
 - partly bound to tannin materials
- Fresh cola seeds are chewed in tropical countries as a stimulant
- vast quantities of dried seeds are processed for the preparation of cola drinks
e.g. Coca-Cola and Pepsi-Cola





Cocoa

- Cocoa (or cacao) is derived from the roasted seeds of *Theobroma cacao* (Sterculiaceae),
- The fruits develop on the trunk of the tree
 - The seeds are separated, allowed to ferment,
 - Then roasted to develop the characteristic chocolate flavour
 - The kernels are then separated from the husks, ground up,
 - Processed in various ways to give chocolate, cocoa, and cocoa butter
- Cocoa seeds contain:
 - 35–50% of oil (cocoa butter or theobroma oil)
 - 1–4% **theobromine**
 - 0.2–0.5% caffeine,
 - tannins and volatile oils
- Theobroma oil or cocoa butter is obtained by hot expression from the ground seeds as a whitish solid with a mild chocolate taste
 - It is a valuable formulation aid in pharmacy where it is used as a **suppository base**
 - It contains glycerides of oleic (35%), stearic (35%), palmitic (26%), and linoleic (3%) acids

